

OVERCONFIDENT FORECASTERS AND THE IMPACT OF INFLATION INFORMATION: EVIDENCE FROM A RANDOMIZED SURVEY EXPERIMENT*

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Abstract

Using a long-running randomized treatment in a survey of Italian firms, we ask whether sustained access to recent inflation data changes how managers update forecasts in response to news. The treatment attenuates deviations from full-information rational expectations (FIRE): relative to non-treated managers, treated managers' consensus forecasts underreact less and individual forecasts overreact less. Among competing models, noisy information with overconfidence in private information fits the evidence best. A common signal with the experiment's precision moves the incomplete-information Phillips curve a quarter of the way toward its FIRE counterpart and raises inflation's impact response to cost-push shocks by about 25 percent.

JEL Codes: C53, D83, D84, E31, E37

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1 Introduction

Inflation expectations play an important role in shaping the economic decisions of households and firms. While most macroeconomic models assume full-information rational expectations (FIRE), this assumption has been challenged based on survey data on expectations of firms, households, and professional forecasters. In a seminal study, Coibion and Gorodnichenko (2015) find that on average, inflation forecasts of professional forecasters underreact to news – that is, they adjust their forecasts less than they should in response to new information – and explain this by maintaining rational expectations but departing from the full information assumption (Sims, 2003; Woodford, 2001). More recent work has shown that individual inflation forecasts overreact to news (Bordalo et al., 2020), which cannot be explained by standard models of information frictions. To simultaneously explain sluggishness in average forecasts and excess sensitivity in individual forecasts, the literature has resorted to combining information rigidities with other frictions/biases, such as noisy memory (Sung, 2026), over-extrapolation (Angeletos et al., 2021), and diagnostic expectations (Bordalo et al., 2020).

Despite these advances, a clear consensus on the right model of expectation formation has yet to emerge. Concurrently, a growing body of literature leveraging randomized controlled trials has found that reminding firms of recent macroeconomic data – such as the latest inflation rate – at the time of the interview has a significant impact on the macroeconomic expectations they report (Coibion et al., 2022, 2023). Regarding inflation expectations, this effect is particularly strong in periods of low inflation (Weber et al., 2025), where providing such information also reduces disagreement about future inflation (Bartiloro et al., 2019; Coibion et al., 2019). However, much of this literature relies on newly established surveys, which makes these settings ill-suited to assessing whether sustained access to information improves the efficiency with which agents incorporate new data into their forecasts. These gaps motivate three central questions for this paper. Does enhanced access to information about recent macroeconomic dynamics bring forecasts closer to the rational expectations benchmark? What does this reveal about the underlying model of expectation formation? And what do the resulting forecast moments imply for aggregate inflation dynamics?

We answer these questions using a randomized information experiment in the Bank of Italy’s Survey of Inflation and Growth Expectations (SIGE), a quarterly survey of managers

of Italian firms. Until 2012 Q2, questions about respondents’ views on future consumer price inflation always included a preamble referencing the latest inflation release. Starting in 2012 Q3, however, this preamble was omitted for roughly one-third of the interviewees. In this environment, our strategy involves testing the FIRE hypothesis by regressing firms’ forecast errors on forecast revisions separately for treated and non-treated firms. We estimate these regressions both at the consensus level (as in Coibion and Gorodnichenko, 2015) and at the individual level (as in Bordalo et al., 2020).

We find that non-treated firms exhibit consensus underreaction and individual overreaction (as in previous studies based on professional forecasters’ expectations). By contrast, the treatment drives the two estimated coefficients markedly toward zero. At the consensus level, treated firms display *less underreaction*, which is consistent with standard models of information frictions. At the individual level, we find that compared to untreated firms, treated firms display *less overreaction*, and we show that this allows us to distinguish between competing models that feature departures from individual rationality. Our individual-level finding is not driven by treated firms mechanically anchoring their forecasts to the provided inflation release, by the treatment merely attenuating measurement-error bias or by the information treatment differentially enabling managers to learn about structural features of the inflation process (e.g., persistence or the long-run mean). Notably, this reduction in individual overreaction is not confined to the six-month horizon: it is equally present at the one-year-ahead horizon, where mechanical anchoring to the provided inflation release is far less likely. Overall, our findings suggest that providing timely information on recent inflation at the time of the survey enables firms to update their beliefs more efficiently, moving their forecasts closer to the full-information rational expectations benchmark.

To distinguish among competing models of departures from individual rationality, we embed these mechanisms in a common noisy-information framework and conduct a formal model-selection exercise. Specifically, we start from a noisy-information environment in which each firm observes an idiosyncratic private signal about inflation, while treated firms additionally observe a common public signal. We then incorporate four leading departures from individual rationality in this setting: internal cognitive constraints/noisy memory (Sung, 2026), overconfidence in private information (Daniel et al., 1998), diagnostic expectations (Bordalo et al., 2020), and over-extrapolation (Angeletos et al., 2021). For each mechanism, we derive closed-form expressions for the consensus- and individual-level error-

on-revision coefficients as functions of the model parameters. We then estimate each model’s parameters in an overidentified Generalized Method of Moments (GMM) exercise that targets the treated and non-treated coefficients and the treatment-induced change in forecast dispersion (which disciplines the precision of the public inflation signal) and we assess overall fit using a GMM J-test. Because the number of targeted moments exceeds the number of free parameters, this approach yields a disciplined set of overidentifying restrictions that lets us formally compare competing models.

Our model-selection exercise indicates that a model with *overconfidence in private information* best explains the observed post-treatment reduction in individual-level overreaction. According to this model, excess sensitivity of individual forecasts is due to forecasters overweighting their private signals relative to what rational agents would do. All else equal, the provision of a public signal leads them to place a smaller weight on their private information when forming their forecasts, thereby reducing overreaction.

The overconfidence model outperforms all competing frameworks that we evaluate. Among them, diagnostic expectations is excluded because the diagnostic distortion scales the Kalman gains on *all* signals proportionally (Bordalo et al., 2020), leaving the individual-level coefficient invariant to a public signal – so it cannot account for the post-treatment reduction in overreaction.¹ Of the remaining alternatives, internal cognitive constraints predict too large a reduction in consensus underreaction and too little attenuation of individual overreaction, while over-extrapolation is rejected because matching the non-treated individual coefficient would require an implausibly large gap between true and perceived persistence. We develop these comparisons in Section 5.

We then ask what these forecast moments imply for aggregate inflation dynamics. To do so, we embed the estimated moments in the incomplete-information Phillips-curve framework of Angeletos and Huo (2021), in which dispersed information makes inflation behave like a hybrid Phillips curve with two departures from the full-information benchmark: *myopia*, meaning inflation responds less than fully to expected future inflation, and *anchoring*,

¹Our analysis evaluates diagnostic expectations as modeled in Bordalo et al. (2020), and therefore does not rule out alternative formulations of diagnostic expectations that may be consistent with our evidence. For example, in the smooth diagnostic expectations model of Bianchi et al. (2026), the strength of the diagnostic distortion depends on conditional uncertainty. To the extent that the information treatment reduces uncertainty, this mechanism could generate the attenuation in individual overreaction that we document. We leave a formal assessment of this mechanism to future work.

meaning it remains partly tied to its own past. The exercise proceeds in three steps. First, we use the non-treated forecast moments to discipline the perceived and true noise in firms’ private information, which pins down the baseline myopia and anchoring wedges in the non-treated economy. Second, we use the treated consensus coefficient to identify the precision of a model-equivalent public signal in a fixed-outcome forecasting problem: the treatment changes treated firms’ information sets, but not the aggregate inflation process they forecast. Third, we ask the counterfactual question of how inflation dynamics would change if a signal with that calibrated precision were common information for all price-setting firms. The counterfactual closes roughly one quarter of the myopia and anchoring gaps in the Phillips curve, raises the normalized impact response of inflation to a cost-push shock by about 25%, and reduces the hump-shaped persistence generated by anchoring.

Literature Review. This paper contributes to four strands of the literature. First, we add to work that tests full-information rational expectations (FIRE) using the predictability of forecast errors from forecast revisions. Existing evidence documents departures from FIRE in professional macroeconomic forecasts (Angeletos et al., 2021; Bordalo et al., 2020; Coibion & Gorodnichenko, 2012, 2015; Kohlhas & Walther, 2021), expectations about financial variables such as stock returns, credit spreads, and earnings (Bordalo et al., 2018, 2024; Bouchaud et al., 2019; Greenwood & Shleifer, 2014), and randomized survey experiments (Afrouzi et al., 2023). We contribute by studying firm managers’ inflation forecasts and by using a randomized information intervention to show how access to recent inflation data changes forecast-error predictability at both the consensus and individual levels.

Second, we contribute to the empirical literature that uses information interventions to study expectation formation and economic decisions. Theoretically, sluggish updating is often rationalized by noisy information or rational inattention (Coibion & Gorodnichenko, 2015), possibly with heterogeneous attention (Link et al., 2023), and information provision can alter the effective information set of respondents (Mackowiak & Wiederholt, 2024). Empirically, inflation news affects the expectations of households and firms (Coibion et al., 2022, 2023; Rosolia, 2024; Weber et al., 2025); reminders about recent inflation reduce short-horizon disagreement (Bartiloro et al., 2019; Coibion et al., 2019), while information about the central bank’s inflation target reduces medium- and long-horizon disagreement (Bottone et al., 2022). We extend this literature by asking not only whether information moves forecasts, but whether it moves the updating process closer to the FIRE benchmark.

Third, we relate to work emphasizing overconfidence in private information as a source of forecast anomalies. Daniel et al. (1998) show how investor overconfidence can generate overreaction and return predictability, while Broer and Kohlhas (2024) develop a noisy-information framework in which forecasters overreact to some public signals and underreact to others because they misperceive signal precisions. Most closely related, Adam et al. (2025) argue that professional forecasters are overconfident in their private information. We provide evidence of a similar force among firm managers: non-treated firms display individual-level overreaction, while the public information treatment attenuates this overreaction by reducing the weight placed on private signals. In a formal model-selection exercise, the noisy-information model with overconfidence provides the best fit relative to alternatives based on diagnostic expectations, cognitive noise, or over-extrapolation.

Finally, we contribute to the literature that uses expectations data to discipline macroeconomic models with informational frictions. Angeletos and Huo (2021) show that incomplete information in price setting is observationally equivalent to a hybrid New Keynesian Phillips curve with myopia and anchoring, and that the consensus error-on-revision coefficient disciplines these wedges. We bring this mapping to a randomized information setting: the treatment-induced change in the consensus coefficient identifies the precision of a public signal, which we then use to quantify the general-equilibrium implications of common inflation information.

The remainder of the paper proceeds as follows. Section 2 provides an overview of the SIGE dataset and describes the design of the randomized information treatment. Section 3 details our methodology and reports the primary empirical findings. Section 4 presents targeted robustness checks to evaluate alternative explanations for our main results. Section 5 examines how our empirical evidence aligns with predictions from different theoretical models of expectation formation. Section 6 quantifies the macroeconomic implications of our findings, mapping the estimated forecast moments into general-equilibrium effects on inflation dynamics. Section 7 concludes.

2 Survey data

The dataset we use is the Survey of Inflation and Growth Expectations (SIGE) run by the Bank of Italy since 1999, which surveys a random sample of managers of Italian firms that have more than 50 employees. Initially targeted towards firms in non-financial private services and industry, it expanded in 2013 to include construction firms. Firms are selected to be part of the survey using a stratified design, with strata defined by (i) Sector: industry, non-financial private services, and construction; (ii) Geography: North-West, North-East, Centre, South and Islands; and (iii) Number of Employees: 50-199, 200-999, and 1,000 or more workers. The survey collects qualitative and quantitative data on firms' expectations about inflation, macroeconomic conditions, and their own business outlook - including anticipated prices, demand, investment and hiring plans, and credit availability. Information is gathered primarily through computer-assisted web interviews with the manager at each firm who is most knowledgeable about these topics. The survey also includes repeated questions to ensure consistency over time and occasionally introduces special modules on relevant economic themes. The SIGE is well-suited to our analysis for three reasons: (i) it spans a long time series; (ii) it elicits expectations at multiple horizons, allowing us to construct forecast revisions; and (iii) it embeds a randomized information treatment that provides the most recent inflation release to a subset of firms. We describe each in turn.

From 1999, the SIGE has elicited 1-year-ahead inflation expectations from firm managers. Over time, the questionnaire was expanded to elicit inflation expectations at different horizons: from 2009 Q2, the survey was expanded to elicit firms' year-on-year inflation expectations 2 years from now, from 2010 Q4, the survey was expanded to elicit firms' inflation expectations 6 months ahead², from 2014 Q1, the survey was expanded to elicit firms' year-on-year inflation expectations 4 years from now. This feature is critical for our analysis, as it allows us to construct forecast revisions, defined as the difference between a firm's current forecast for a given horizon and the forecast it made in the past for the same target date. For example, a firm surveyed in period $t - h$ and again in period t provides two forecasts for inflation at time $t + h$, formed at different points in time. The difference between the firm's

²Six-month-ahead year-on-year inflation refers to the YoY rate for the month or quarter occurring six months after the interview. For example, if a firm is surveyed in March 2012, the six-month-ahead YoY rate is the rate of inflation from September 2011 to September 2012.

time t and time $t - h$ forecasts of inflation at $t + h$ captures how new information observed between $t - h$ and t is reflected in its updated forecast.

Information treatment. A unique feature of the SIGE survey is the inclusion of a randomized information treatment, introduced in 2012 Q3. Before this date, all surveyed firms were provided with information about the most recent realization of inflation in Italy before being asked to report their inflation expectations. Beginning in 2012 Q3, firms were randomly assigned to one of two groups. The control group comprising roughly 33% of the sample received no such information and were simply asked to report their inflation expectations across multiple horizons. In contrast, the treated group continued to receive information about the most recent 12-month inflation rate in Italy. Due to publication lags, this corresponds to the year-on-year inflation rate observed two months before the date of the survey. Treatment assignments are persistent: firms placed in the treatment group continue receiving information on recent inflation until the next reshuffling. These reshufflings occurred in 2012 Q4, 2017 Q2, and 2019 Q4.

To ensure data quality, at each horizon, we restrict the sample to firms who provided at least ten forecasts in the sample, and exclude forecasts that are more than five interquartile ranges away from that horizon’s median. Figure 1 plots the resulting distribution of 6-month-ahead inflation forecasts across six different survey quarters, separately for treated and non-treated firms. The plot reveals that treated firms’ forecast distributions tend to shift closer to the provided inflation rate (indicated by a vertical dashed line) while also exhibiting lower cross-sectional dispersion relative to the control group. This suggests that the information treatment may have an effect on expectations and reduce disagreement across firms (also shown in Table 2). Appendix Figure A.1 replicates the same structure as Figure 1 but focuses on 1-year ahead inflation expectations. As with the 6-month horizon, treated firms’ expectations are more tightly clustered around the provided inflation figure. Relative to the one-year horizon, treated firms’ six-month-ahead inflation forecasts cluster more tightly around the provided benchmark, consistent with firms viewing recent inflation as a better predictor of future inflation at shorter horizons.

Table 1 contrasts treated and non-treated firms across some of the variables reported in SIGE, such as inflation expectations, firm characteristics, and beliefs. Treated firms report higher inflation expectations, with the treated-control gap narrowing as the forecast horizon

lengthens. Consistent with higher inflation expectations, they also expect to make larger price changes over the next 12 months. Otherwise, the two groups appear similar across a broad set of SIGE covariates, including assessments of current economic conditions and own business situation, export intensity, and the perceived sensitivity of business conditions to demand and to future credit conditions.

3 Empirical Analysis

3.1 Model

To empirically test the full-information rational expectations (FIRE) hypothesis in our setting, we rely on the error-on-revision regressions at the consensus level (Coibion & Gorodnichenko, 2015) and individual level (Bordalo et al., 2020). The consensus-level regression is as follows

$$x_{t+h} - x_{t+h|t} = \alpha_C + \beta_C(x_{t+h|t} - x_{t+h|(t-h)}) + \eta_{t+h|t} \quad (3.1)$$

where the dependent variable is the average h -quarter-ahead forecast error (constructed as the difference between actual and expected inflation) and the main regressor is the h -quarter revision in the average inflation forecast for quarter $t + h$, i.e., the difference between the average forecast formed at quarter t and the one formed earlier at $t - h$. Similarly, the individual-level regression has the following form:

$$x_{t+h} - x_{t+h|t}^i = \alpha_i + \beta_I(x_{t+h|t}^i - x_{t+h|(t-h)}^i) + \eta_{t+h|t}^i \quad (3.2)$$

where the dependent variable is the h -quarter ahead individual forecast error and the main independent variable is the h -quarter revision in the individual forecast.³ We estimate equation (3.2) with firm fixed effects to absorb time-invariant heterogeneity - for example, a firm manager's persistent optimism or pessimism. To maintain consistency with the literature that uses professional forecaster data, our baseline forecast horizon is $h = 2$ (six months).⁴

³Here $x_{t+h|t}^i$ denotes the forecast of x_{t+h} made by firm i at time t and $x_{t+h|t} = \int x_{t+h|t}^i$ denotes the corresponding average forecast

⁴Coibion and Gorodnichenko (2015) and Bordalo et al. (2020) use a one-quarter revision window, $x_{t+h|t} - x_{t+h|t-1}$ and a baseline horizon of $h = 3$ quarters.

Thus, in equation (3.1) the dependent variable is the average six-month-ahead forecast error, while the independent variable is the difference between the average six-month-ahead inflation forecast at quarter t and the average one-year ahead inflation forecast formed at quarter $t - 2$. Relative to the large literature that studies deviations from FIRE by examining the predictability of forecast errors from forecast revisions, our specifications differ only in using a two-quarter (rather than a one-quarter) revision window. Since the independent variables in equations (3.1) and (3.2) are known at time t , FIRE implies that the coefficients β_C and β_I should be zero⁵. We estimate both specifications separately by treatment status to assess how the information intervention shapes systematic departures from rational expectations. Because the six-month revision spans two survey waves, a firm's treatment status can differ at its two endpoints; in the individual-level regressions we therefore retain only revisions for which the firm's status is the same at t and $t - 2$, excluding the small share of revisions that straddle a reshuffling of the treatment assignment. As noted in Section 2, our sample begins in 2010 Q4 – the first quarter in which six-month-ahead inflation forecasts were collected. Finally, to focus on typical business conditions and avoid the regime shift associated with the post-2021 inflation surge, we end the sample in 2020 Q4.⁶

To test directly whether the treatment shifts the error-on-revision coefficient, we also estimate both regressions on the pooled sample of treated and non-treated firms. At the consensus level,

$$\begin{aligned} x_{t+h} - x_{t+h|t} = & \alpha_C + \delta T + \beta_C^{NT} (x_{t+h|t} - x_{t+h|(t-h)}) \\ & + \beta_{DIFF}^C T (x_{t+h|t} - x_{t+h|(t-h)}) + \eta_{t+h|t} \end{aligned} \quad (3.3)$$

and at the individual level,

$$\begin{aligned} x_{t+h} - x_{t+h|t}^i = & \alpha_{i,T_i} + \beta_I^{NT} (x_{t+h|t}^i - x_{t+h|(t-h)}^i) \\ & + \beta_{DIFF} T_i (x_{t+h|t}^i - x_{t+h|(t-h)}^i) + \eta_{t+h|t}^i \end{aligned} \quad (3.4)$$

where T (resp. T_i) is a treatment indicator equal to one for treated observations and zero otherwise. In the consensus regression, the dummy δT lets the treated and non-treated series

⁵To align with the empirics, the theoretical analysis in Section 5 indexes time in six-month periods; hence the two-quarter revision used here corresponds to a one-period revision in the model in Section 5.

⁶This choice is consistent with recent evidence that information treatments are most informative in low-inflation environments (Weber et al., 2025).

differ in their average forecast error; in the individual regression, the firm-by-treatment fixed effects α_{i,T_i} – a separate intercept for each firm’s treated and non-treated spells – absorb the same level shift. The non-treated slope is β^{NT} , and the treatment-induced difference $\beta_{DIFF} = \beta(T) - \beta(NT)$, reported in the “Pooled” column of each table, measures how far the treatment moves the coefficient. Under FIRE – or if the public signal leaves updating unchanged – $\beta_{DIFF} = 0$.

Table 2 reports summary statistics on our main dependent and independent variables for treated and non-treated firms. Panel A reports statistics for the consensus forecast errors and forecast revisions, and Panel B reports statistics for individual forecast errors and forecast revisions. Panel A shows that while both treated and non-treated firms forecast higher inflation relative to realized inflation, non-treated firms make larger forecast errors. This pattern also appears in Figure 2, which plots realized inflation alongside average 6-month- and 1-year-ahead forecasts for treated firms (panel a) and non-treated firms (panel b). Moreover, note that the consensus forecast errors and forecast revisions of treated firms are more dispersed than those of non-treated firms.

Panel B of Table 2 reports statistics on individual-level forecasts. First, we observe that there is more dispersion in the forecasts of non-treated firms⁷, in line with what we show in Figure 1. We define the non-revision share as the fraction of firm-quarter observations in which the current six-month-ahead inflation forecast differs from the one-year-ahead forecast made two quarters earlier by less than 0.01 percentage points. This statistic is a measure of rigidity in forecasts with respect to new information, and is used as a measure of forecaster attentiveness in the literature (Andrade & Le Bihan, 2013). By this metric, treated firms exhibit slightly lower but overall similar rigidity compared with non-treated firms, suggesting that the two groups are not significantly different in terms of attentiveness to macroeconomic developments. We also report average spell length by treatment status – the mean number of quarters a firm is observed as treated (and analogously as non-treated). Because the control group only stopped receiving the information treatment in 2012 Q3, treated spells are slightly longer on average. Finally, we observe that the individual forecast errors of non-treated firms are more dispersed than those of treated firms, while the standard deviation of individual forecast revisions is roughly the same for treated and non-treated firms.

⁷We define forecast dispersion as the average cross-sectional standard deviation of 6-month ahead inflation forecasts for treated and non-treated firms.

3.2 Results

Table 3 reports our main results from estimating equations (3.1) and (3.2) for the treated and non-treated firms. First, we observe that non-treated firms (columns 2 and 5 of Table 3) exhibit underreaction at the consensus level and overreaction at the individual level, mirroring the evidence obtained from professional forecasters’ expectations (Bordalo et al., 2020). Firms that receive the treatment show a lower consensus coefficient β_C , while the value of the individual coefficient β_I is less negative. This suggests that information treatments can help firms incorporate new information into their forecasts in a manner that is more consistent with the FIRE benchmark. Another noteworthy feature of Table 3 is the sharp decline in R-squared values when equations (3.1) and (3.2) are estimated for treated firms. This implies that forecast errors are less predictable from forecast revisions among treated firms, consistent with the idea that the information treatment improves forecast efficiency and brings expectations closer to the rational benchmark.

Pooling the two groups makes the comparison explicit. The Pooled columns of Table 3 estimate the difference specifications (3.3) and (3.4). At the consensus level the treatment lowers the slope by $\beta_{DIFF}^C = -0.82$ (Newey-West $p = 0.03$), about two-thirds of the non-treated coefficient of 1.25; at the individual level it raises the slope by $\beta_{DIFF} = 0.14$ ($p = 0.04$; wild-cluster bootstrap $p = 0.04$), enough to close essentially the entire gap between the non-treated overreaction of -0.28 and the treated value of -0.15 . Both differences are significant at conventional levels. We report β_{DIFF} and its p -value for every specification that follows; where the difference is imprecise, we also ask whether the treated coefficient is itself distinguishable from zero – that is, whether treated firms already forecast rationally.

Our baseline individual-level specification includes firm fixed effects, so β_I is identified from within-forecaster variation in revisions and absorbs time-invariant heterogeneity in forecasting behavior. Appendix B.6 reports the corresponding estimates without firm fixed effects. The non-treated coefficient remains negative and statistically significant, while the treated coefficient is statistically indistinguishable from zero; the pooled treated–non-treated difference is smaller and imprecisely estimated.

Overall, the information treatment moves both the consensus- and individual-level error-on-revision coefficients toward the FIRE benchmark. While the consensus evidence is consistent with standard models of information frictions (as formalized in Section 5), the

treatment-induced attenuation of individual overreaction provides key evidence for distinguishing among competing models of deviations from individual rationality. Accordingly, Section 4 addresses two design-specific alternative explanations for the individual-level evidence—learning about the inflation process and experimenter demand effects—while Appendix B.1 addresses measurement-error concerns. Section 5 then embeds the consensus and individual moments in a model-selection exercise.

4 Robustness Checks

Our setting and empirical strategy raise two design-specific concerns for interpreting the individual-level results. First, because treatment assignment is persistent, treated firms may gradually learn structural features of the inflation process, such as persistence or the long-run mean, rather than simply using the public inflation signal more efficiently. Second, treated firms may mechanically anchor their responses to the inflation release displayed in the survey, so that measured forecasts partly reflect experimenter-demand effects rather than underlying beliefs. We address these two concerns in the main text below. A separate, standard econometric concern is that measurement error in reported forecasts may mechanically generate negative individual error-on-revision coefficients; Appendix B.1 addresses this issue using cross-horizon revisions that break the direct overlap between the dependent variable and the revision. Across these exercises, the main interpretation is unchanged.

4.1 Learning About the DGP

Farmer et al. (2024) show that the forecast anomalies documented in the literature can arise even under Bayesian updating, when agents do not know the underlying data-generating process (DGP) and learn about parameters that characterize the long-run properties of the DGP. They also demonstrate that learning converges more quickly when agents face fewer unknowns. In our context, treated firms consistently receive information about recent inflation over the sample period. As a result, treated firms may gradually develop a better understanding of structural features of the DGP - such as inflation persistence, or the long-run mean - which could lead to a reduction in forecast anomalies over time.

To test this hypothesis, we leverage another unique feature of the SIGE treatment design. As described in Section 3, prior to 2012 Q3, all surveyed firms received information about the most recent inflation release in Italy. Beginning in 2012 Q3, firms assigned to the control group were drawn from this previously informed pool. We therefore focus on the early years of the treatment rollout - when non-treated firms had only recently lost access to the information treatment after several quarters of receiving it. Specifically, we restrict the sample to the period from 2012 Q4 to 2015 Q4 and re-estimate equation (3.2) for both treated and non-treated firms. Table 4 presents the results from this exercise alongside our baseline estimates from Table 3. We find that, if anything, non-treated firms in this restricted window exhibit even stronger individual-level overreaction, suggesting that our main results are not driven by treated firms learning structural features of the DGP (e.g., persistence or the long-run mean) more accurately due to the information treatment. In this early window the pooled difference is, if anything, larger than in the full sample ($\beta_{DIFF} = 0.25$, $p < 0.01$; wild-cluster bootstrap $p < 0.01$). A treatment effect that strengthens precisely when the control group has only just lost access to the information is hard to square with gradual learning about the data-generating process (DGP).

4.2 Experimenter Demand Effects

Treated firms may anchor their inflation forecasts to the most recent realization provided as part of the treatment, rather than reporting their underlying beliefs. This concern is especially relevant for 6-month-ahead forecasts, where firms may reasonably interpret recent inflation as a highly informative signal about near-term inflation dynamics. To gauge the severity of this issue, we first report statistics on the number of firms in the treated and non-treated samples that report the most recent inflation realization as their forecast at different horizons. At the 6-month horizon, approximately 15% of treated firm forecasts exactly match the provided recent inflation figure, compared to only 6% for non-treated firms. At the 12-month horizon, these proportions fall to 9% for treated firms and 4% for non-treated firms, which shows that mechanically anchoring forecasts to the provided inflation release is higher at shorter horizons. Even fewer firms (6% treated, 1.7% non-treated) align forecasts exactly at both horizons.

To address this issue, we pursue two approaches. First, we estimate our baseline re-

gression at a longer forecast horizon. Specifically, we regress the 1-year-ahead forecast error on the forecast revision defined as the difference between the current 1-year-ahead inflation forecast and the 2-year-ahead forecast made four quarters earlier. Since inflation is less persistent at longer horizons, recent realizations are less directly informative, thereby reducing the likelihood that treated firms mechanically anchor their forecasts to the provided inflation figure (as shown above). Table 5 presents the results from these long-horizon individual-level regressions, alongside the baseline results from Table 3. The similarity of the coefficients across both horizons for treated and non-treated firms suggests that our main findings are not driven by experimenter demand effects. Importantly, this exercise shows that the attenuation of individual overreaction is itself present at the longer horizon, not only in the baseline: at the one-year horizon the treated individual coefficient is again markedly less negative than the non-treated one (-0.15 versus -0.28), exactly mirroring the six-month estimates. The pooled difference is correspondingly 0.14 ($p = 0.04$) at six months and a similar 0.13 ($p = 0.09$) at one year, so lengthening the horizon – and weakening any anchoring to the provided release – leaves the individual-level attenuation roughly intact.

Second, in our baseline regressions, we exclude firms in the treated sample that report both their 6-month and 1-year-ahead inflation forecasts to be exactly equal to the inflation figure provided to them as part of the treatment as this subset of firms is more likely to be mechanically anchoring their forecasts. Column (3) of Table B.4 presents the results from this specification. We find that the estimated β_I coefficient remains very similar to our baseline estimate for the treated sample, providing additional evidence that experimenter demand effects are not driving our main results. Comparing this trimmed treated sample with non-treated firms, the pooled difference is $\beta_{DIFF} = 0.14$ ($p = 0.03$; wild-cluster bootstrap $p = 0.03$), essentially identical to the baseline. Dropping the firms most likely to anchor mechanically thus leaves the result unchanged.

5 Theoretical framework

In this section, we assess how well existing models of expectation formation can account for the new empirical facts documented in Section 3. The inflation process, x_t , follows an AR(1). Across all model variants, firms observe idiosyncratic private signals about x_t . We model

the survey's information intervention as giving treated firms access to an additional public signal about recent inflation at the time forecasts are formed. The models differ in how they generate excess sensitivity of individual forecasts. Specifically, we evaluate the following leading mechanisms that feature departures from individual rationality - internal cognitive constraints (Sung, 2026), diagnostic expectations (Bordalo et al., 2020), overconfidence in private information (Daniel et al., 1998), and over-extrapolation (Angeletos et al., 2021). For each mechanism, we provide closed-form expressions for the consensus and individual-level error-on-revision regression coefficients and evaluate model fit using a J-test of overidentifying restrictions. Detailed derivations are present in Appendix C.

Throughout this section, we maintain the following assumptions about the underlying process x_t and the signal structure. A continuum of forecasters $i \in [0, 1]$ forecast x_{t+h} for $h \in \{0, 1, \dots, H\}$. Its current value x_t follows an AR(1) process:

$$x_t = \rho x_{t-1} + u_t ; \quad u_t \sim N(0, \sigma_u^2) \quad (5.1)$$

where $u_t \sim N(0, \sigma_u^2)$. We assume that the agents are aware of the long-run mean, which we normalize to 0, as well as the parameters ρ and σ_u^2 . The current value of x_t is not perfectly observed, and each forecaster i instead observes a noisy signal:

$$s_t^i = x_t + \epsilon_{it}; \quad \epsilon_{it} \sim N(0, \sigma_\epsilon^2) \quad (5.2)$$

Treated firms, in addition, also have access to an informative public signal:

$$s_t = x_t + \nu_t; \quad \nu_t \sim N(0, \sigma_\nu^2) \quad (5.3)$$

For non-treated firms, we assume that the public signal is shut down; i.e., $\sigma_\nu^2 \rightarrow \infty$. At each time t , forecasters minimize the expected quadratic loss from inaccurate projections across H horizons:

$$\sum_{h=0}^H (x_{t+h} - F_{i,t}(x_{t+h}))^2 \quad (5.4)$$

This implies that the forecast of x_{t+h} for agent i is $\rho^h F_{i,t}(x_t)$, where $F_{i,t}(x_t)$ is agent i 's

best guess of the current value of the hidden state variable, which we denote as $x_{t|t}^i$. Then, $F_{i,t}(x_{t+h}) = x_{t+h|t}^i = \rho^h x_{t|t}^i$. Also, we denote $x_{t+h|t} = \int_0^1 x_{t+h|t}^i$. We now derive the regression coefficients that are the theoretical counterparts of the empirical specifications in equations (3.1) and (3.2).

5.1 Internal Cognitive Constraints

This section adapts the framework in Section 2 of Sung (2026), but instead of modeling endogenous information acquisition, we assume that agents receive exogenous noisy signals about the underlying state, as specified in equations (5.2) and (5.3). While agents are assumed to know the process parameters, they face cognitive costs in retrieving information from previous periods, which increases their sensitivity to new information and generates overreaction at the individual level.

Let m_{it} denote the mental representation of existing knowledge at the start of period t , while $M_{it} = (m_{it-1}, s_{it-1}, s_{t-1})$ denotes the prior knowledge that is available to the agent before she observes the noisy signals at time t . Following Sung (2026), we assume that the mental representation of existing knowledge takes the following form:

$$m_{it} = \Lambda_t M_{it} + \tilde{\omega}_{it}; \quad \tilde{\omega}_{it} \sim N(0, \Sigma_{\omega t}) \quad (5.5)$$

The agent faces a constraint on the mutual information between m_{it} and M_{it} :

$$\mathcal{I}(m_{it}, M_{it}) \leq -\frac{\log \phi_m}{2} \quad (5.6)$$

In this case, Sung (2026) shows that the optimal memory representation takes the following form:

$$m_{it} = \lambda_t \mathbb{E}[x_t | M_{it}] + \omega_{it}; \quad \omega_{it} \sim N(0, \sigma_{\omega t}) \quad (5.7)$$

where $\lambda_t \in [0, 1]$ and $\mathbb{E}[x_t | M_{it}] = x_{t|t-1}^i$. In Appendix C.1, we show that under these assumptions, $\lambda_t = \lambda = 1 - \phi_m$. In general, there are multiple values of $\sigma_{\omega t}^2$ and λ that satisfy the constraint (5.6). Therefore, we impose the following normalization that allows us to

determine $\sigma_{\omega t}^2$ purely as a function of λ :

$$\text{Cov}[\mathbb{E}[x_t|M_{it} - m_{it}, m_{it}]] = 0 \quad (5.8)$$

This allows us to derive the following expressions for the asymptotic limit of β_C and β_I (a detailed exposition is in Appendix C.1):

Proposition 1. *When forecasters form expectations subject to the Noisy Memory constraint and receive both idiosyncratic private signals and a public signal, the asymptotic error-on-revision coefficients are*

$$\beta_I = - \frac{(1-K)(1-\lambda)}{\rho^{-2} + 1 - 2[(1-K)\lambda + K]} \quad (5.9)$$

and

$$\beta_C = \frac{1-K}{K} - \frac{K_2^2 \sigma_\nu^2}{K V_{agg}} + \frac{(1-K)(1-\lambda)}{K} \frac{C_{x_{t|t-1}, agg}}{V_{agg}}. \quad (5.10)$$

where K_1 and K_2 denote the steady state Kalman gains on the private and public signals, respectively, with $K = K_1 + K_2$, V_{agg} denotes the variance of the aggregate forecast revision and $C_{x_{t|t-1}, agg}$ denotes the covariance between $x_{t|t-1}$ and the aggregate forecast revision $\square$

As long as $\lambda < 1$, it follows that $\beta_I < 0$. Intuitively, limits on recalling past information lead forecasters to place more weight on newly observed signals, which in turn generates individual-level overreaction. In the absence of a public signal, we also have $\beta_C > 0$. However, as shown by Coibion and Gorodnichenko (2015), introducing a common signal can bias β_C downward (given by the third term in 5.10) and may even result in negative values of β_C . Details on how we calculate $C_{x_{t|t-1}, agg}$ and V_{agg} are in Appendix C.1.

5.2 Diagnostic Expectations

Bordalo et al. (2020) develop a model of expectation formation in which agents receive idiosyncratic noisy signals and exhibit belief distortions driven by Kahneman and Tversky's representativeness heuristic that accounts for both consensus underreaction and individual-level overreaction. The representativeness of a state x at time t is defined by the following

likelihood ratio:

$$R_t(x) = \frac{f(x | S_t^i)}{f(x | S_{t-1}^i \cup \{x_{t-1}^i\})}, \quad (5.11)$$

where S_t^i represents the complete set of signals observed by agent i up to time t (for treated firms, S_t^i includes both public and private signals), and $f(x | S_t^i)$ denotes the rational posterior belief about the state x after observing these signals. The forecaster then overweights representative states by using the following distorted posterior:

$$f^\theta(x_t | S_t^i) = f(x_t | S_t^i) R_t(x_t)^\theta \frac{1}{Z_t} \quad (5.12)$$

where Z_t is a normalizing factor that ensures that $f^\theta(x_t | S_t^i)$ integrates to 1. θ governs the extent to which forecasters overweight the absolute probability of states that have become relatively more likely after observing today's signals. Under these assumptions, we have the following expressions for the asymptotic limit of β_C and β_I :

Proposition 2. *Under Diagnostic Expectations, when forecasters receive both idiosyncratic private signals and a public signal, the asymptotic error-on-revision coefficients are*

$$\beta_I = -\frac{\theta(1+\theta)}{(1+\theta)^2 + \theta^2 \rho^2} \quad (5.13)$$

and

$$\beta_C = \frac{\left(\frac{1-K}{K} - \theta\right) ([1+\theta - \theta\rho^2(1-K)]V_{agg}^R + \theta\rho^2 K_2^2 \sigma_\nu^2) - \frac{(1+\theta)K_2^2 \sigma_\nu^2}{K}}{V_{agg}} \quad (5.14)$$

where K_1 and K_2 denote the steady state Kalman gains on the private and public signals, respectively, with $K = K_1 + K_2$, V_{agg} is the variance of the aggregate forecast revision under diagnostic expectations and V_{agg}^R is the variance of the forecast revision under rational expectations $\square$

A detailed exposition of this proposition is provided in Section A of the Online Appendix of Bordalo et al. (2020) and is therefore omitted here. As shown in equation (5.13), the individual-level error-on-revision regression coefficient in the diagnostic expectations model depends only on the diagnostic distortion θ and the process persistence ρ . Thus, while the provision of a public signal affects β_C , it leaves β_I unchanged. Consequently, the diagnos-

tic expectations framework of Bordalo et al. (2020) cannot account for the post-treatment increase in β_I documented in Section 3 ⁸. Intuitively, the diagnostic distortion θ scales the Kalman gains on *all* signals proportionally, which implies that forecasters overreact to both public and private signals.

5.3 Overconfidence

We now consider a model in which forecasters are overconfident about the precision of their private signals (Broer & Kohlhas, 2024; Daniel et al., 1998). While the true variance of the idiosyncratic signal is σ_ϵ^2 , agents believe it to be $\hat{\sigma}_\epsilon^2 < \sigma_\epsilon^2$; that is, they perceive their private signals as more informative than they actually are. We assume that forecasters correctly assess the informativeness of the public signal, as defined in equation (5.3). Under these assumptions, we have the following expressions for the coefficients β_C and β_I .

Proposition 3. *When forecasters observe idiosyncratic private signals as well as a public signal and are overconfident about the precision of their private signal, the asymptotic error-on-revision coefficients are*

$$\beta_I = \frac{1 - K}{K} - \frac{K_1^2 \sigma_\epsilon^2}{K V_{ind}} - \frac{K_2^2 \sigma_\nu^2}{K V_{ind}}. \quad (5.15)$$

and

$$\beta_C = \frac{1 - K}{K} - \frac{K_2^2 \sigma_\nu^2}{K V_{agg}}. \quad (5.16)$$

where K_1 and K_2 denote the steady state (distorted) Kalman gains on the private and public signals, respectively, with $K = K_1 + K_2$, V_{ind} denotes the variance of individual forecast revisions and V_{agg} denotes the variance of the aggregate forecast revisions. $\square$

Because agents overestimate the precision of their private signals, they update more aggressively than a rational agent would in response to any given realization of that signal. This excessive updating generates systematic overreaction, and yields a negative value of β_I .

⁸A natural extension is the Smooth Diagnostic Expectations model of Bianchi et al. (2026), in which the severity of the diagnostic distortion varies with conditional uncertainty. If the information treatment lowers conditional uncertainty, smooth DE would predict the rise in β_I that we observe empirically. We leave a full analysis to future work

Expressions for the Kalman gains and the variance of the forecast revisions are provided in Appendix C.2.

5.4 Over-extrapolation

In this model, we assume that while the true data-generating process continues to follow equation (5.1), agents believe that the process is more persistent than it actually is, i.e., they believe the underlying state evolves according to:

$$x_t = \hat{\rho}x_{t-1} + u_t; \quad u_t \sim N(0, \sigma_u^2) \quad (5.17)$$

The asymptotic error-on-revision regression coefficients are then given by the following expressions:

Proposition 4. *When forecasters have extrapolative expectations and observe idiosyncratic private signals as well as a public signal, the asymptotic error-on-revision coefficients are*

$$\beta_I^h = \frac{1-K}{K} - \frac{K_1^2 \sigma_\epsilon^2}{K V_{ind}} - \frac{K_2^2 \sigma_\nu^2}{K V_{ind}} + \frac{(\rho^h - \hat{\rho}^h) C_{x,Ind}}{\hat{\rho}^h V_{ind}}. \quad (5.18)$$

and

$$\beta_C^h = \frac{1-K}{K} - \frac{K_2^2 \sigma_\nu^2}{K V_{agg}} + \frac{(\rho^h - \hat{\rho}^h) C_{x,agg}}{\hat{\rho}^h V_{agg}}. \quad (5.19)$$

where K_1 and K_2 denote the steady state (distorted) Kalman gains on the private and public signals, respectively, with $K = K_1 + K_2$. Here, V_{ind} and V_{agg} denote the variances of the individual and aggregate forecast revisions, respectively, and $C_{x,Ind}$ and $C_{x,agg}$ denote the covariances of x_t with the individual and aggregate forecast revisions, respectively. $\square$

Since agents overestimate the persistence of the underlying process, they expect shocks to x_t to have more prolonged effects than they truly do, leading to negative ex-post forecast errors and a negative β_I . Among the models considered in this section, the over-extrapolation model is unique in that the coefficients β_C^h and β_I^h depend on the forecast horizon h .⁹ This

⁹This holds for the Noisy Memory model only because we assume a known long-run mean, so that the h -period-ahead forecast is $\rho^h x_{t|t}^i$ and the horizon scaling cancels. When the long-run mean is itself a latent state, as in Section 3 of Sung (2026), the memory system prioritizes storing this low-frequency component,

is evident from equations (5.18) and (5.19). The reason is that in this model, agents make forecast errors because $\hat{\rho} \neq \rho$, and the magnitude of these errors varies with the forecast horizon h . Expressions for the Kalman gains, variances of the individual and aggregate forecast revisions and covariance of x_t with the individual and aggregate forecast revisions are given in Appendix C.3.

5.5 Analysis

We now assess the ability of the three models – internal cognitive constraints, overconfidence in private information, and over-extrapolation – to quantitatively match the empirical findings documented in Section 3.¹⁰ To this end, we implement a model selection exercise by testing overidentifying restrictions using a J -test. We begin by estimating the parameters of the data-generating process, ρ and σ_u^2 , using historical data on consumer price inflation in Italy. We then specify the free parameters associated with each model and describe the moments used in our test of overidentifying restrictions. Finally, we present the results of the test, showing that the overconfidence model provides the best fit, with additional intuition and just-identified exercises reported in Appendix D.1.

First, we fit an AR(1) process to the de-meaned quarterly series of year-on-year changes in the Harmonized Index of Consumer Prices (HICP) for Italy¹¹. This allows us to estimate the persistence of the process, denoted ρ_q , and the variance of its innovations, denoted $\sigma_{u,q}^2$. To ensure consistency with the two-quarter revision window used in our empirical analysis, we transform these estimates using the relationships $\rho = \rho_q^2$ and $\sigma_u^2 = (1 + \rho_q^2)\sigma_{u,q}^2$. These parameters are treated as fixed throughout the analysis, although we examine the robustness of our results to several alternative specifications of the data-generating process in Appendix D.

Each model features three free parameters: the variance of the private signal, σ_ϵ^2 ; a model-specific behavioral parameter $\hat{\sigma}_\epsilon^2$ in the overconfidence model, λ in the internal cognitive constraints model, and $\hat{\rho}$ in the over-extrapolation model; and the variance of the public

long-horizon forecasts lean more on imperfect recall, and both β_C and β_I become more negative with the horizon.

¹⁰Recall that we excluded the diagnostic expectations model since the expression for β_I in that model is independent of σ_ν^2

¹¹This is the inflation rate that firms in our survey are asked to forecast.

signal for treated firms, σ_ν^2 . Model-implied non-treated moments are obtained by evaluating the same closed-form expressions at $\sigma_\nu^2 \rightarrow \infty$ (i.e., zero public signal precision). Our test of overidentifying restrictions targets five empirical moments. The first four are the consensus and individual error-on-revision regression coefficients for non-treated firms, $\beta_C(NT)$ and $\beta_I(NT)$, and for treated firms, $\beta_C(T)$ and $\beta_I(T)$. The fifth moment is the ratio of the average cross-sectional standard deviation of six-month-ahead inflation forecasts for treated firms relative to non-treated firms, which disciplines the value of σ_ν^2 . Intuitively, the availability of a common public signal causes all firms to update based on the same signal realization, thereby reducing forecast dispersion in theory and in the data (see Table 2). Analytical expressions for the cross-sectional variance of forecasts under different model specifications are given in Appendix C.

5.6 Overidentification test

We estimate the parameters of the three models using an overidentified Generalized Method of Moments (GMM) procedure and assess model fit using a J-test. Let $\hat{m}$ denote the empirical moments and $m(\theta)$ denote the model-implied moments for a given parameter vector θ . For each model, we choose θ to solve the following minimization problem:

$$\hat{\theta} = \arg \min_{\theta} (m(\theta) - \hat{m})' W (m(\theta) - \hat{m}), \quad (5.20)$$

where W , the optimal weighting matrix, is the inverse of the variance-covariance matrix of the five moments that we consider. To estimate the optimal weighting matrix, we implement a fixed-length time-block bootstrap. In each bootstrap draw, we sample, with replacement, blocks of twenty consecutive quarters to assemble a bootstrap time path that is the same length as our original panel, and we retain all firms observed in the selected quarters. We recompute the five moments on this resampled panel and use the sample variance-covariance matrix across bootstrap draws as our estimate of the optimal variance matrix. Because we have five moments and three parameters, the minimized value of the GMM objective function in equation (5.20) is distributed as a chi-squared random variable with two degrees of freedom. The parameter estimates from this procedure are given in Table 6, and the estimated model fit and J-statistic from our GMM exercise are reported in Table 7.

First, note that the forecast dispersion ratio is the most precisely estimated moment in the bootstrap, so it receives the greatest weight in the optimal matrix and effectively disciplines σ_ν^2 across specifications. As a result, all three models match this moment fairly well. Among the alternatives, the Overconfidence (OC) model provides the best fit. It reproduces the non-treated coefficients β_C and β_I quite well, and for the implied value of σ_ν^2 it generates changes in the treated coefficients that are of comparable magnitude to those observed in the data. The resulting J -statistic is small, with an associated p -value of 0.91, indicating that the overidentifying restrictions are comfortably satisfied. Taken together, these results point to a strong overall fit of the OC model relative to the other specifications.

In contrast, the Noisy Memory model reproduces the non-treated coefficients reasonably well, but—given σ_ν^2 that is disciplined by the dispersion ratio—it implies a much larger drop in β_C for treated firms (0.03 vs. 0.43 in the data) and only a modest increase in β_I (-0.25 vs. -0.15 in the data). The resulting J -statistic is higher and the p -value is significantly lower ($p = 0.15$): we do not reject at conventional levels, but the fit is notably weaker than the OC model, with this model failing to match both treated coefficients for reasonable values of σ_ν^2 .

Finally, the over-extrapolation model delivers the weakest fit out of the alternatives we consider. Since coefficients in this framework are horizon-dependent, we set $h = 1$ (corresponding to a six-month horizon, consistent with our empirical analysis in Section 3). To generate $\beta_I(NT) = -0.28$ at this short horizon, the model requires $\hat{\rho}$ to be substantially larger than the true persistence parameter ρ . However, with the restriction $\hat{\rho} < 1$, the model implied $\beta_I(NT)$ is much closer to zero (-0.15 in our calibration) relative to what we observe in the data. It also generates a counterfactually large value of β_C for the non-treated firms. As a result, the J -statistic is large and the associated p -value falls below 1% ($p < 0.01$), leading to a strong rejection of the model’s overidentifying restrictions.

The model-comparison results have a simple interpretation. In the Overconfidence model, individual overreaction is generated by forecasters placing too much weight on private signals. A public signal therefore operates through two margins at once: it speeds up aggregate updating, reducing β_C , and it reduces the relative weight placed on private information, pushing β_I toward zero. This joint movement is exactly what we observe among treated firms.

The alternative models struggle because the public signal does not directly undo the force that generates individual overreaction. In the Noisy Memory model, weak priors make agents highly responsive to current signals, so a public signal can generate a large decline in β_C . But the memory friction remains: only a fraction of current information is carried into the next period, so β_I moves only modestly. Forcing the model to match the treated β_I would require an extremely precise public signal, which would collapse forecast dispersion and imply a counterfactually low treated β_C . In the Over-extrapolation model, individual overreaction comes from perceived persistence $\hat{\rho}$ rather than signal noise, so increasing public-signal precision does little to correct β_I . Appendix D.1 reports just-identified exercises that make these tradeoffs explicit. They lead to the same conclusion as the overidentified GMM test: overconfidence is the only specification that can jointly match the treated and non-treated moments without implausible public-signal precision or counterfactual forecast dispersion.

6 Macroeconomic Effects of Information Provision

In Section 3, we show that information treatments move firm managers' belief updating closer to what full-information rational expectations (FIRE) implies. In Section 5, we show that a model featuring noisy information and overconfidence in private information best explains these findings. This section asks what those forecast moments imply for equilibrium aggregate outcomes. We answer this question within the framework of Angeletos and Huo (2021), henceforth A&H. In their inflation application, firms set prices on the basis of dispersed, noisy signals of aggregate real marginal cost. The resulting equilibrium dynamics are observationally equivalent to a hybrid New Keynesian Phillips curve (NKPC) with myopia in the forward-looking term and anchoring to past inflation. This framework is useful for our purposes because it provides a disciplined mapping from forecast error-on-revision moments – in particular the consensus coefficient of Coibion and Gorodnichenko (2015) (β_C), to the Phillips-curve wedges that govern aggregate inflation dynamics. We adapt the A&H environment by allowing firms to be overconfident in their private signals, in line with our findings in Section 5.

Our counterfactual exercise proceeds in three steps. First, we use our estimates of the

consensus and individual error-on-revision regressions, β_C and β_I , to pin down the perceived and true levels of noise in firms' private signals.¹² The perceived noise in private information then pins down the myopia and anchoring wedges in the baseline, non-treated economy. Second, we model the information treatment as a model-equivalent public signal of aggregate real marginal cost and discipline its precision using the consensus error-on-revision coefficient estimated in the treated sample. Importantly, in this step aggregate inflation dynamics are held fixed at their baseline non-treated law of motion. This restriction reflects the experimental setting: the treatment is provided to a randomized subset of survey respondents and therefore improves treated firms' forecasts without changing the aggregate inflation process they forecast. The public signal thus affects beliefs, forecast revisions, and forecast errors in the identification step, but not equilibrium aggregate outcomes. Third, after identifying the precision of the public signal in this fixed-outcome forecasting problem, we ask what would happen if a signal of that same precision were made available to all price-setting firms as common information. This final step maps the calibrated signal through the A&H public-signal equilibrium and therefore allows for the full general-equilibrium effects of common information on price setting, higher-order beliefs, and aggregate inflation dynamics.

The implied effects are sizable. The common-information counterfactual closes roughly one quarter of the myopia and anchoring gaps in the Phillips curve, raises the impact response of inflation to a normalized cost-push shock by about 25%, and reduces the hump-shaped persistence generated by anchoring.

6.1 Environment and the non-treated calibration

The incomplete-information NKPC. The firms' price setting problem follows from the inflation application of A&H. A continuum of firms set prices subject to a Calvo friction; a firm resets its price each period with probability $1 - \theta$ and discounts the future at rate

¹²In A&H, the key moment that disciplines the myopia and anchoring wedges is the consensus error-on-revision coefficient, β_C . Conditional on matching this moment, allowing firms to be overconfident in their private signals changes individual-level moments, such as β_I and forecast dispersion, but does not change equilibrium aggregate outcomes. Intuitively, β_C identifies perceived private noise, which governs the A&H root and therefore the Phillips-curve wedges; the gap between perceived and true private noise disciplines overconfidence.

χ .¹³ Firms observe noisy private signals about real marginal cost and form expectations by Bayesian updating. Aggregating reset prices delivers the incomplete-information NKPC

$$\pi_t = \kappa \sum_{k \geq 0} (\chi\theta)^k \bar{\mathbb{E}}_t[mc_{t+k}] + \chi(1-\theta) \sum_{k \geq 0} (\chi\theta)^k \bar{\mathbb{E}}_t[\pi_{t+k+1}], \quad (6.1)$$

where $\bar{\mathbb{E}}_t$ denotes the average expectation across firms, mc_t is real marginal cost, and κ is the slope of the Phillips curve (Angeletos & Huo, 2021, eq. 25 and Online Appendix D). Under FIRE, (6.1) collapses to the textbook NKPC. Away from FIRE, current inflation depends on the entire term structure of average forecasts of marginal costs and of future inflation. The second sum is the general-equilibrium (GE) component: firms' beliefs about other firms' future price setting decisions, and its weight $\chi(1-\theta)$ rises with the frequency of price adjustment.

Fundamental and information. Real marginal cost follows an AR(1),

$$mc_t = \rho mc_{t-1} + u_t, \quad u_t \sim N(0, \sigma_u^2), \quad (6.2)$$

with one model period equal to six months, consistent with the two-quarter revision window of Section 3.¹⁴ We measure mc_t by the Italian labor income share and set $\rho = 0.78$ at the six-month frequency.¹⁵ The calibration of the CG moments, the friction roots, and the NKPC wedges depends on signal precision relative to the innovation variance of the fundamental. We therefore follow A&H and normalize $\sigma_u^2 = 1$; shock magnitudes in the subsequent New Keynesian impulse responses are normalized separately.¹⁶

¹³Throughout this section θ denotes the Calvo no-reset probability, following A&H; it is unrelated to the diagnostic-expectations parameter of Section 5.2.

¹⁴Section 5 treats inflation itself as the exogenous state being forecast, the appropriate abstraction for comparing models of belief formation. Here inflation is an equilibrium object, so persistence belongs to the fundamental; we calibrate ρ directly to the empirical proxy for marginal cost, as in Galí and Gertler (1999), Galí et al. (2005), and A&H.

¹⁵We estimate quarterly AR(1) processes on the log labor share of the Italian total economy (compensation of employees over gross value added, adjusted for self-employment as in Gollin (2002)), 1995Q1–2019Q4. Following the convention of A&H, who report estimates with and without a linear trend and choose an intermediate value, we take the midpoint of our two estimates (0.8900 without and 0.8731 with a trend) and convert it to the six-month model period by squaring, which yields $\rho = 0.7771$.

¹⁶We accordingly express all signal variances and precisions below in units of the fundamental's innovation variance. All targeted and reported moments (β_C , β_I , dispersion ratios), the friction roots, and the NKPC

Information is dispersed, as in Section 5, but the object of learning differs. In Section 5, inflation itself is the exogenous state being forecast. In the A&H inflation economy, inflation is instead an equilibrium outcome, and the primitive state that firms learn about is aggregate real marginal cost. In the baseline, each firm i observes a private signal of this fundamental,

$$s_t^i = mc_t + \epsilon_{it}, \quad \epsilon_{it} \sim N(0, \sigma_\epsilon^2). \quad (6.3)$$

In the treatment calibration and counterfactual exercises below, treated firms in addition observe a common public signal,¹⁷

$$z_t = mc_t + \nu_t, \quad \nu_t \sim N(0, \sigma_\nu^2). \quad (6.4)$$

Firms form expectations by Bayesian updating on the history of the signals they observe. We allow for firms to be overconfident in their private information, in line with the specification selected by the model selection exercise of Section 5.6: they perceive the variance of their private noise to be $\hat{\sigma}_\epsilon^2 < \sigma_\epsilon^2$, while they perceive the precision of the public signal, $\tau_\nu \equiv \sigma_\nu^{-2}$, correctly. A result of A&H (their Proposition 7) makes the overconfidence layer tractable: *all aggregate outcomes depend on the information structure only through the perceived noise $\hat{\sigma}_\epsilon^2$* ; the true noise σ_ϵ^2 matters only for individual-level moments such as β_I and the cross-sectional dispersion of forecasts. The aggregate quantification below therefore runs on $\hat{\sigma}_\epsilon^2$, while β_I separately disciplines σ_ϵ^2 .¹⁸

Equilibrium and observational equivalence. A&H show that this economy has a unique equilibrium (their Proposition 2): the friction dampens the response of inflation

wedges are dimensionless and invariant to this normalization.

¹⁷The mapping from the survey treatment to this signal is model-equivalent rather than literal. In the survey, treated firms receive the latest published inflation release. In the A&H environment, however, inflation is an equilibrium outcome driven by aggregate real marginal cost. We therefore represent the treatment as a noisy common signal about the inflation-relevant fundamental. This is the parsimonious way to carry the experimental information into the A&H model: information about recent inflation is informative about the marginal-cost state that generates non-treated equilibrium inflation. A&H also consider both a public signal of the fundamental and a noisy public statistic of inflation and show that the former provides a tractable analytical stand-in for the latter.

¹⁸This invariance is special to overconfidence: under the other departures considered in Section 5, the behavioral parameter alters equilibrium dynamics even conditional on β_C .

to fundamentals on impact and propagates it through an endogenous root ϑ ,

$$\pi_t = A q_t, \quad q_t = \vartheta q_{t-1} + m c_t, \quad A = \left(1 - \frac{\vartheta}{\rho}\right) \frac{\kappa}{1 - \rho\chi}, \quad \vartheta \in (0, \rho), \quad (6.5)$$

where ϑ summarizes the information friction: it is the reciprocal of the unique outside root of a cubic given in Appendix E.3, it vanishes under FIRE, and it rises with noise and with the strength of the GE feedback. Their observational-equivalence result (Proposition 3 and Corollary 2) then states that the dynamics (6.5) are identical to those generated by a full-information *hybrid* NKPC,

$$\pi_t = \kappa m c_t + \chi \omega_f \mathbb{E}_t[\pi_{t+1}] + \omega_b \pi_{t-1}, \quad (6.6)$$

with $\omega_f < 1$ (*myopia*: extra discounting of the future) and $\omega_b > 0$ (*anchoring*: backward-looking indexation-like behavior). The pair (ω_f, ω_b) is not free: both wedges are explicit functions of the single root,

$$\omega_f = \frac{\chi \rho^2 - \vartheta}{\chi (\rho^2 - \vartheta^2)}, \quad \omega_b = \frac{\vartheta (1 - \chi \vartheta) \rho^2}{\rho^2 - \vartheta^2} \quad (6.7)$$

(Angeletos & Huo, 2021, Online Appendix, eqs. A5–A6). More noise and a stronger GE feedback both push ω_f down and ω_b up (their Propositions 4,5); under FIRE ($\vartheta = 0$), $\omega_f = 1$ and $\omega_b = 0$.

The remaining two blocks of the model are standard. Because our empirical evidence disciplines belief formation only on the price-setting side – the error-on-revision regressions of Section 3 are estimated on firms’ inflation forecasts – the information friction enters through the NKPC alone, and households and the monetary authority are as in the full-information model. We therefore close the economy with a FIRE IS curve and a Taylor rule,

$$\begin{aligned} \tilde{y}_t &= \mathbb{E}_t[\tilde{y}_{t+1}] - \sigma_c^{-1} (i_t - \mathbb{E}_t[\pi_{t+1}]), \\ i_t &= \phi_\pi \pi_t + \phi_y \tilde{y}_t, \end{aligned} \quad (6.8)$$

with output gap $\tilde{y}_t$, inverse EIS $\sigma_c = 1$, and Taylor coefficients $\phi_\pi = 1.5$ and $\phi_y = 0.25$.

Calibration. The NKPC parameters are those of A&H, time-aggregated from quarterly to the six-month model period (Table 8): $\chi = 0.99^2 = 0.98$, $\theta = 0.60^2 = 0.36$. We fix the slope κ at 0.17 in all three information regimes and calibrate the information friction independently of it, following A&H, who treat κ as a free scale parameter.¹⁹ We calibrate the information parameters to the survey moments of Section 3, mirroring the model of Section 5.3 inside the equilibrium economy:

$$\beta_C(NT) = 1.25 \Rightarrow \hat{\sigma}_\epsilon^2 = 2.16, \quad \beta_I(NT) = -0.28 \Rightarrow \sigma_\epsilon^2 = 4.68. \quad (6.9)$$

The first step uses the closed-form expression for the consensus error-on-revision coefficient in the private-information equilibrium (Angeletos & Huo, 2021, eq. 24), which is monotone in the perceived noise; the second step matches the individual coefficient (Bordalo et al., 2020), computed under the calibrated perceived noise but with signals drawn at the true noise (Appendix E.1). The implied overconfidence ratio, $\kappa_{oc} \equiv \hat{\sigma}_\epsilon^2 / \sigma_\epsilon^2 = 0.46$, says that firms perceive their private information to be roughly twice as precise as it truly is – in line with the GMM estimates of Section 5.6. The resulting non-treated wedges are

$$\vartheta_{NT} = 0.53, \quad \omega_f^{NT} = 0.19, \quad \omega_b^{NT} = 0.48 :$$

the consensus underreaction documented in Section 3 leaves the Phillips curve with one-fifth of its frictionless forward-looking weight and an anchoring weight of nearly one-half on past inflation.

6.2 Calibrating the precision of the information treatment

The remaining parameter is the fixed-outcome partial-equilibrium (PE) precision, τ_ν^{PE} , of the model-equivalent public signal implied by treated firms' forecasts. Here the design of the experiment motivates the identification protocol. Treated firms are a randomized, atomistic subset of the forecasters in the survey: the treatment changes what they know, but it does not change the inflation process that the Italian economy generates – information shown to a minority of forecasters inside a survey does not affect aggregate inflation. The treated

¹⁹Under our Calvo frequency conversion, $\kappa = 0.17$ corresponds to a quarterly slope on real marginal cost of about 0.04, in line with the GMM estimates of Galí and Gertler (1999); Appendix E provides details.

coefficient $\beta_C(T)$ therefore measures how much better treated firms forecast the unchanged, non-treated equilibrium outcome. We impose exactly this structure: in the identification step, the public signal affects beliefs only, never the equilibrium law of motion.

Formally, let π_t^{NT} denote non-treated equilibrium inflation – the law (6.5) evaluated at the non-treated root ϑ_{NT} of Section 6.1. In the identification step, a treated firm forecasts π_{t+1}^{NT} using both signals in (6.3) and (6.4) – its private signal and the model-equivalent treatment signal – while the process π_t^{NT} itself is held fixed.²⁰ The model analogue of β_C is the population error-on-revision coefficient at the one-period (six-month) horizon:

$$K_{CG}^{PE}(\tau_\nu) = \frac{\text{Cov}(FE_t, REV_t)}{\text{Var}(REV_t)}, \quad \begin{aligned} FE_t &= \pi_{t+1}^{NT} - \bar{\mathbb{E}}_t[\pi_{t+1}^{NT}], \\ REV_t &= \bar{\mathbb{E}}_t[\pi_{t+1}^{NT}] - \bar{\mathbb{E}}_{t-1}[\pi_{t+1}^{NT}], \end{aligned} \quad (6.10)$$

where $\bar{\mathbb{E}}_t$ averages the treated firms' posterior means. We compute the coefficient from the stationary distribution of the joint state and average-belief dynamics; Appendix E.2 gives the state-space representation of the fixed-outcome system, develops the computation, and shows that the private-information closed form of A&H obtains at $\tau_\nu = 0$. A more precise common signal makes average beliefs update faster and consensus errors less predictable, so K_{CG}^{PE} is decreasing in τ_ν . Matching the treated coefficient gives

$$K_{CG}^{PE}(\tau_\nu) = \beta_C(T) = 0.43 \quad \implies \quad \tau_\nu^{PE} = 0.21. \quad (6.11)$$

The implied *public-signal share of perceived precision* is

$$\psi^{PE} \equiv \frac{\tau_\nu^{PE}}{\hat{\sigma}_\epsilon^{-2} + \tau_\nu^{PE}} = 0.31. \quad (6.12)$$

Thus, in normalized A&H units, the treatment identifies a signal that accounts for roughly 30% of perceived precision about the forecast-relevant state. This precision is identified only from the treated consensus coefficient; treated individual overreaction and forecast dispersion

²⁰In the survey, the treatment is the latest published inflation release. In the model, we represent it as an equivalent common signal about the forecast-relevant state: non-treated inflation is driven by marginal cost through (6.5), so information about recent inflation is, through the lens of the model, information about the marginal-cost process that generates it. Angeletos and Huo (2021) consider both a public signal of the fundamental and a noisy public statistic of inflation itself (their Online Appendix D) and show that the former is a good analytical stand-in for the latter.

are reported as diagnostics, not targets. Evaluated in the A&H public-signal counterfactual at the calibrated precision, the model implies a treated individual coefficient of about -0.21 , compared with -0.15 in the data, and a treated/non-treated forecast-dispersion ratio of about 0.92 , compared with 0.70 . Both moments move in the empirically observed direction, but neither is used to identify τ_ν^{PE} ; forecast dispersion in particular is never targeted in the macro calibration.²¹

6.3 The general-equilibrium counterfactual

Having measured the precision of the public signal from the consensus error-on-revision coefficient of treated firms' forecasts, we now study the consequences of a signal with the calibrated precision being made common information to all price-setting firms in the economy – a stylized counterpart of communication policies that make inflation-relevant information salient and commonly observed. Relative to the identification step, two things change. First, every firm's information improves. Second, the signal becomes *common information*: every firm knows that other firms also observe it, which sharpens higher-order beliefs and therefore the GE component of the NKPC (6.1). A&H characterize the resulting equilibrium in the Online Appendix (Proposition 12): the friction root ϑ now solves a generalization of their cubic in which total information enters through the combined precision $\hat{\sigma}_\epsilon^{-2} + \tau_\nu^{PE}$, while the *publicness* of the signal enters only through the GE weight $\chi(1 - \theta)$, discounted by the share ψ^{PE} of (6.12), which now measures the commonly observed fraction of firms' perceived precision (Appendix E.3). Holding total precision fixed, the distinction between private and common information would be irrelevant without the GE component. Here, commonness matters because pricing decisions feed back into one another, so the signal also sharpens higher-order beliefs. The common-information counterfactual is also motivated by survey evidence that information treatments can move firms' beliefs about other firms' beliefs, not only their own forecasts (Coibion et al., 2021).

Table 9 reports the result. The common-information counterfactual moves both wedges toward FIRE: the forward-looking multiplier rises from 0.19 to 0.39 , and the anchoring weight

²¹The public-signal counterfactual evaluated at τ_ν^{PE} is the environment that gives the precision its structural interpretation; the diagnostics are moments of that environment, not of the fixed-outcome identification step.

falls from 0.48 to 0.36. Each wedge closes about 24% of its distance to the FIRE benchmark. The effective coefficient on expected inflation, $\chi\omega_f$, doubles.

How much of this movement reflects better information, and how much the commonness of the signal? The cubic of Appendix E.3 separates the two: total precision enters it directly, whereas publicness enters only through the GE weight $\chi(1 - \theta)$, discounted by the share ψ . We therefore re-solve the cubic at the treated total precision, $\hat{\sigma}_\epsilon^{-2} + \tau_\nu^{PE}$, in two steps. In the first, we set $\psi = 0$: every firm observes a signal of the same total precision, but privately, so higher-order beliefs are unchanged. In the second, we restore $\psi^{PE} = 0.31$ and make the signal common. Better information alone lowers the friction root from 0.530 to 0.461 and closes 18% of each wedge's gap to the FIRE benchmark; making the signal common lowers the root further to 0.434 and closes the remaining 6 percentage points. Roughly three quarters of the counterfactual's effect on the Phillips-curve wedges is thus the value of better information, and one quarter the value of common knowledge.

Macroeconomic outcomes. To translate wedges into dynamics, we embed the hybrid NKPC (6.6) in a textbook three-equation New Keynesian model at the six-month frequency, closing it with the FIRE IS curve and Taylor rule (6.8), and add AR(1) cost-push and monetary disturbances with six-month persistence $\rho_s = 0.81$.²² Figure 3 plots the responses of inflation, the output gap, and the nominal rate to a one-standard-deviation cost-push shock under the three regimes; the monetary-policy shock, with the same message, is in Appendix E.4. Because the shock size is a normalization, the object of interest is how the responses differ across information regimes rather than their level.

Under complete information, inflation jumps on impact and decays monotonically. Under the non-treated calibration, the information friction cuts the impact response of inflation by more than half (from 2.12 to 0.92) and produces the characteristic hump: the peak response arrives only in the second half-year after the shock. Information provision moves the economy toward the FIRE benchmark. The impact response of inflation rises by about 25% (from 0.92 to 1.15), closing roughly a fifth of the gap to complete information on impact. The peak still arrives in the second half-year, but the hump flattens relative to its own impact

²²The cost-push disturbance enters the hybrid NKPC (6.6) additively and the monetary disturbance enters the Taylor rule; both are distinct from the marginal-cost innovation u_t of (6.2). Only (ω_f, ω_b) differ across the three regimes.

and inflation decays faster as a share of its peak at medium horizons – both movements toward the monotone FIRE response. Better information thus makes inflation respond more promptly to the same disturbance while reducing the hump-shaped persistence generated by anchoring. The output gap and the nominal rate inherit the same pattern through the policy rule.

Interpretation. The counterfactual should not be read as the general-equilibrium effect of the survey treatment itself. It quantifies how inflation dynamics would change in the A&H model if all price setters received a common signal with the precision identified from the survey treatment.

7 Conclusion

Using a randomized information experiment embedded in the Bank of Italy’s Survey of Inflation and Growth Expectations, we study how timely information about recent inflation affects firms’ belief updating. Non-treated firms display the same pair of forecast anomalies documented in other expectations data: consensus forecasts underreact to news, while individual forecasts overreact. Providing firms with the latest inflation reading moves both coefficients toward the FIRE benchmark. Treated firms exhibit less consensus underreaction and less individual overreaction, and their forecast errors become less predictable from their revisions.

We then use the experiment to distinguish among competing models of expectation formation. A noisy-information model with overconfidence in private signals provides the best overall account of the evidence. In this model, firms perceive their private signals to be more precise than they truly are. As a result, they place too much weight on private information, generating individual-level overreaction. The public signal reduces this overweighting of private information, while also speeding up consensus updating. Alternative mechanisms, including noisy memory, diagnostic expectations, and over-extrapolation, cannot jointly match the treated and non-treated consensus and individual moments as successfully.

Finally, we quantify the macroeconomic implications of these forecast moments. Embedding the evidence in the incomplete-information Phillips-curve framework of Angeletos and

Huo (2021), we use the non-treated moments to discipline the baseline myopia and anchoring wedges and the treated consensus coefficient to identify the precision of an equivalent public signal in a fixed-outcome forecasting problem. Feeding that calibrated precision into the A&H public-signal equilibrium, we find that making such a signal common to all price setters steepens the short-run response of inflation to shocks and dampens the persistence generated by anchoring.

These macroeconomic results should be interpreted as a counterfactual, not as the general-equilibrium effect of the survey treatment itself. The treatment is administered to a randomized subset of survey respondents and therefore changes treated firms' information sets, not the aggregate inflation process. The counterfactual instead asks how inflation dynamics would change if all price-setting firms received a common signal with the precision identified from the experiment. The results suggest that clear and salient inflation information can matter not only for the accuracy of firms' forecasts, but also for the higher-order-belief channel through which information frictions generate inflation persistence.

Our evidence comes from a single country, a single forecast outcome, and a pre-2021 sample; whether these patterns generalize to other settings and to higher-inflation regimes remains to be tested.

Table 1: Summary Statistics: Treated vs. Non-Treated Firms

Statistic Treatment Status	N. Obs		Mean		SD	
	No	Yes	No	Yes	No	Yes
<i>Variable:</i>						
Inflation Forecast (6-Months)	8,580	21,217	0.99	1.11	0.88	1.07
Inflation Forecast (1-Year)	8,580	21,217	1.14	1.22	0.89	1.05
Inflation Forecast (2-Year)	8,580	21,217	1.34	1.37	0.93	1.02
Log Number of Employees	8,580	21,217	5.34	5.42	1.32	1.41
Beliefs about current macroeconomic conditions	8,462	20,946	1.85	1.83	0.60	0.62
Improvement of macroeconomic conditions in 3 months	8,549	21,141	1.88	1.88	0.90	0.86
Share of turnover from exports	8,580	21,217	2.02	2.01	1.13	1.11
Expected price change next 12 months	8,580	21,217	0.72	0.78	3.70	4.20
Beliefs about own business conditions	8,418	20,859	3.43	3.46	0.79	0.78
Effect of Changes in Demand on Business Conditions	6,853	17,441	0.34	0.38	1.59	1.57
Effect of Credit Availability and Cost of Credit on Business Conditions	8,439	20,775	0.01	-0.02	1.37	1.40

Table 1 reports summary statistics for treated and non-treated firms in the SIGE dataset across a range of variables, such as inflation expectations at different horizons, number of employees (log), share of turnover from exports, changes in prices they expect to make over the next 12 months, optimism about future business conditions and factors affecting their business conditions over the next 3 months.

Table 2: Forecast Errors and Revisions: Treated vs. Non-Treated Firms

Panel A: Consensus Errors and Revisions

Variable	Mean (Treated)	Mean (Non-treated)	SD (Treated)	SD (Non-treated)	SE (Treated)	SE (Non-treated)
Forecast Error (6 Months)	-0.23	-0.49	0.82	0.71	0.21	0.21
Forecast Revision (6 Months - lagged 1 Year)	-0.19	-0.26	0.51	0.24	0.12	0.07

Panel B: Individual-Level Forecast Statistics

Variable	Treated	Non-treated
Forecast Dispersion	0.47	0.67
Non-revision share	0.31	0.35
Avg no. of obs per firm	18.4	13.24
Standard Deviation- Forecast Errors	0.88	0.99
Standard Deviation- Forecast Revisions	0.73	0.72

Note: Panel A of Table 2 reports summary statistics on consensus forecast errors and revisions for the 6-month-ahead inflation forecast. Forecast errors are computed as the difference between realized inflation and the consensus forecast, while forecast revisions are defined as the difference between the current 6-month-ahead forecast and the 1-year-ahead forecast made two quarters prior. Columns (1) and (2) report the time-series average of these statistics for treated and non-treated firms, respectively. Columns (3) and (4) report the time-series standard deviation, and columns (5) and (6) report Newey-West standard errors. Panel B of Table 2 reports summary statistics for individual-level forecasts. Forecast Dispersion is the average within-quarter standard deviation of individual 6-month-ahead forecasts. Non-revision share is the fraction of firms whose current 6-month-ahead forecast is within 0.01% of their 1-year-ahead forecast from two quarters earlier. Avg. no. of obs. per firm indicates the average number of periods a firm appears in the treated and non-treated groups, respectively. The final two rows report the standard deviations of individual forecast errors and forecast revisions over time.

Table 3: Full Sample Estimates of β_C and β_I

Dependent Variables: Sample:	Consensus Error (6 Months)			Individual Error (6 Months)		
	Treated	Non-treated	Pooled	Treated	Non-treated	Pooled
β_C	0.43** (0.21)	1.25*** (0.30)				
β_C^{NT}			1.25*** (0.30)			
β_{DIFF}^C			-0.82** (0.37)			
β_I				-0.15* (0.08)	-0.28*** (0.06)	
β_I^{NT}						-0.28*** (0.06)
β_{DIFF}						0.14** (0.06)
<i>Fixed-effects</i>						
firm				Yes	Yes	
firm $\times$ treatment						Yes
<i>Fit statistics</i>						
Observations	39	34	73	15,266	5,805	21,071
R ²	0.08832	0.30668	0.19582	0.18723	0.32974	0.23589
Within R ²				0.01734	0.05621	0.02872

Table 3 presents the results from estimating equations (3.1) and (3.2). Columns (1)–(3) report the consensus-level error-on-revision regression (3.1); standard errors are Newey-West with automatic bandwidth selection. Columns (4)–(6) report the individual-level regression (3.2) with firm fixed effects; standard errors are clustered by firm and by time. The Pooled columns (3) and (6) report the difference specifications (Equations (3.3) and (3.4)): β^{NT} is the non-treated coefficient and β_{DIFF} the treated–non-treated difference. The individual difference is $\beta_{DIFF} = 0.14$ ($p = 0.040$; wild-cluster bootstrap $p = 0.044$); the consensus difference is $\beta_{DIFF}^C = -0.82$ ($p = 0.032$). Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 4: β_I Estimates: Robustness to Learning Dynamics

Dependent Variable: Sample:	Individual Forecast Error (6 Months)					
	Treated	Non-treated	Pooled	Treated	Non-treated	Pooled
β_I	-0.15* (0.08)	-0.28*** (0.06)		-0.14* (0.07)	-0.39*** (0.03)	
β_I^{NT}			-0.28*** (0.06)			-0.39*** (0.03)
β_{DIFF}			0.14** (0.06)			0.25*** (0.06)
<i>Fixed-effects</i>						
firm	Yes	Yes		Yes	Yes	
firm $\times$ treatment			Yes			Yes
<i>Fit statistics</i>						
Observations	15,266	5,805	21,071	4,917	2,306	7,223
R ²	0.18723	0.32974	0.23589	0.31650	0.58009	0.49255
Within R ²	0.01734	0.05621	0.02872	0.03305	0.22579	0.11509

Estimates of (3.2) for the full sample (columns 1–3) and an early treatment window, 2012 Q4–2015 Q4 (columns 4–6), with firm fixed effects. The treated–non-treated difference is significant in both: $\beta_{DIFF} = 0.14$ ($p = 0.040$) full-sample and 0.25 ($p = 0.002$) in the early window. Standard errors clustered by firm and quarter.

Table 5: β_I Estimates: Long Horizon

Dependent Variables: Sample:	Forecast Error (6M)			Forecast Error (12M)		
	Treated	Non-treated	Pooled	Treated	Non-treated	Pooled
β_I	-0.15* (0.08)	-0.28*** (0.06)				
β_I^{NT}			-0.28*** (0.06)			
β_{DIFF}			0.14** (0.06)			
β_I (12 Months)				-0.15 (0.09)	-0.28*** (0.05)	
β_I^{NT} (12 Months)						-0.28*** (0.05)
β_{DIFF} (12 Months)						0.13* (0.07)
<i>Fixed-effects</i>						
firm	Yes	Yes		Yes	Yes	
firm $\times$ treatment			Yes			Yes
<i>Fit statistics</i>						
Observations	15,266	5,805	21,071	13,767	4,936	18,703
R ²	0.18723	0.32974	0.23589	0.21926	0.34582	0.25414
Within R ²	0.01734	0.05621	0.02872	0.01123	0.03358	0.01672

Estimates of (3.2) using 6-month-ahead (columns 1–3) and 12-month-ahead (columns 4–6) forecast errors; at the 12-month horizon the regressor is the 1-year-ahead minus the 2-year-ahead forecast from $t - 4$. The treated–non-treated difference is $\beta_{DIFF} = 0.14$ ($p = 0.040$) at six months and 0.13 ($p = 0.087$) at one year, so the effect persists at longer horizons. Standard errors clustered by firm and quarter.

Table 6: Calibration Details: Overidentification Test

Parameter	Model		
	Overconfidence	Noisy Memory	Extrapolation
ρ (Fix)	0.85	0.85	0.85
σ_u^2 (Fix)	0.42	0.42	0.42
σ_ϵ^2	1.84	1.21	2.40
$\hat{\sigma}_\epsilon^2$	0.85	—	—
λ	—	0.50	—
$\hat{\rho}$	—	—	0.97
σ_ν^2 (Treated)	1.31	1.75	3.96

Table 6 reports the values of our parameters used in the Overidentified GMM Test

Table 7: Results: Overidentification Test

Moment	Data	Model		
		Overconfidence	Noisy Memory	Extrapolation
β_C (T)	0.43	0.33	0.03	0.25
β_I (T)	-0.15	-0.18	-0.25	-0.12
β_C (NT)	1.25	1.21	1.08	1.65
β_I (NT)	-0.28	-0.28	-0.28	-0.15
SD ratio (T/NT)	0.70	0.71	0.73	0.71
J-statistic		0.19	3.78	14.63
p-value		0.91	0.15	<0.01

Table 7 reports the fit of all three models from the Over-identified GMM Test.

Table 8: Calibration

Parameter		Value	Source / target
<i>Panel A: Fixed macro parameters</i>			
ρ	persistence of mc_t	0.7771	Italian labor share, 1995Q1–2019Q4
σ_u^2	innovation variance	1	normalization (Angeletos–Huo)
χ	discount factor	0.9801	0.99 ² , Angeletos–Huo (2021)
θ	Calvo no-reset probability	0.36	0.60 ² , Angeletos–Huo (2021)
κ	NKPC slope	0.17	Galí–Gertler (1999), freq.-adjusted
σ_c	inverse EIS	1	standard
ϕ_π, ϕ_y	Taylor coefficients	1.5, 0.25	standard
ρ_s	disturbance persistence	0.81	0.90 ² quarterly
<i>Panel B: Information parameters identified from survey moments</i>			
$\hat{\sigma}_\epsilon^2$	perceived private noise	2.16	$\beta_C(NT) = 1.25$
σ_ϵ^2	true private noise	4.68	$\beta_I(NT) = -0.28$
κ_{oc}	overconfidence ratio	0.46	$\hat{\sigma}_\epsilon^2 / \sigma_\epsilon^2$
τ_ν^{PE}	public-signal precision	0.21	$\beta_C(T) = 0.43$
ψ^{PE}	public-signal share of precision	0.31	$\tau_\nu^{PE} / (\hat{\sigma}_\epsilon^{-2} + \tau_\nu^{PE})$

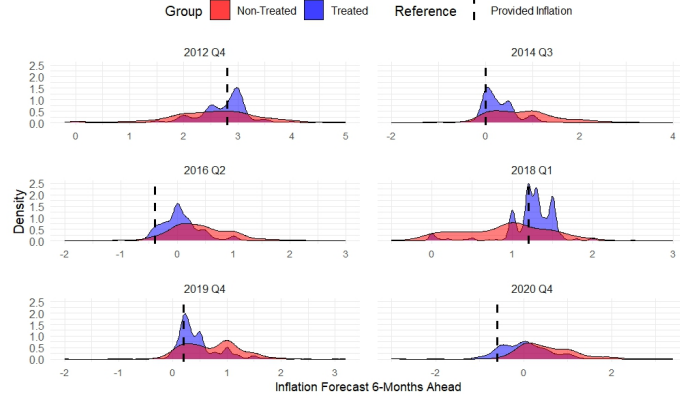
Notes: One model period is six months. EIS denotes the elasticity of intertemporal substitution. Signal variances and precisions are in units of the fundamental innovation variance, normalized to $\sigma_u^2 = 1$. The information parameters in Panel B are calibrated sequentially from the survey moments; κ_{oc} and ψ^{PE} are scale-free.

Table 9: Information frictions and NKPC wedges across regimes

Regime	ϑ	ω_f	ω_b	$\chi \omega_f$
Complete information (FIRE)	0	1	0	0.98
Non-treated	0.530	0.194	0.477	0.190
Treated total precision, signal kept private	0.461	0.341	0.390	0.334
Treated counterfactual	0.434	0.387	0.363	0.379

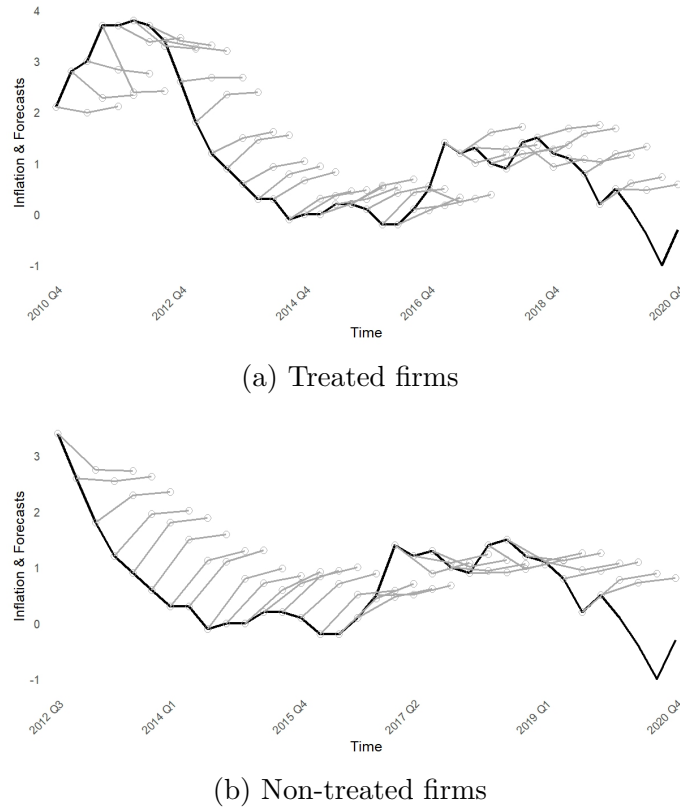
Notes: ϑ is the equilibrium friction root; ω_f and ω_b are the hybrid-NKPC wedges in (6.6); $\chi \omega_f$ is the effective coefficient on expected inflation. The treated counterfactual endows all price setters with a common signal of precision $\tau_\nu^{PE} = 0.21$. The intermediate row is the first step of the decomposition in Section 6.3, not a separate calibration: every firm observes a private signal of the same total precision $\hat{\sigma}_\epsilon^{-2} + \tau_\nu^{PE}$, with the public share ψ set to zero.

Figure 1: Distribution Of 6-Months Ahead Inflation Forecasts



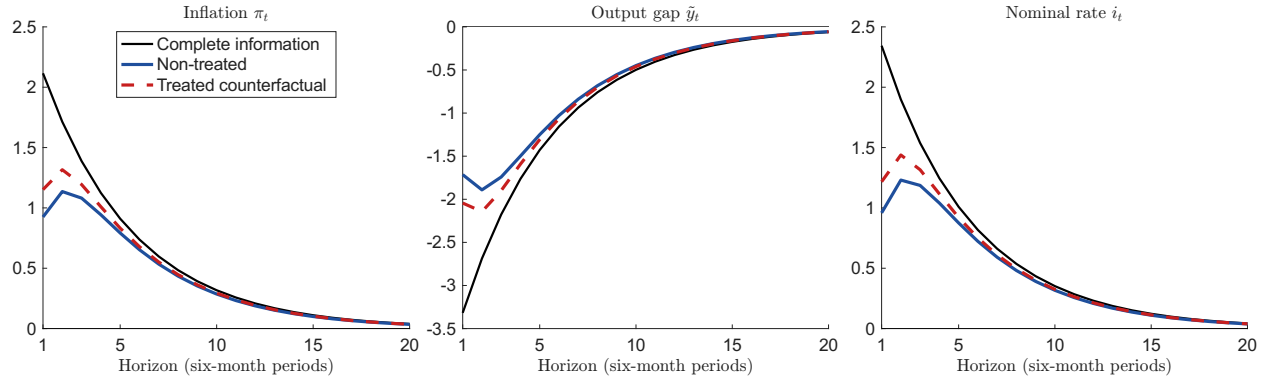
Kernel density estimates of 6-month-ahead inflation expectations for treated and non-treated firms. The vertical dashed line indicates the most recent 12-month inflation rate provided to treated firms at the time of the survey. Each panel corresponds to a different survey quarter.

Figure 2: Inflation Rate and Average Inflation Forecasts: Treated and Non-Treated Firms



Quarterly realized inflation (solid black) alongside mean 6- and 12-month-ahead inflation forecasts (gray lines with circle markers) across the sample period. Panel (a): treated firms; panel (b): non-treated firms.

Figure 3: Responses to a cost-push shock under the three information regimes



Impulse responses of inflation, the output gap, and the nominal interest rate to a one-standard-deviation cost-push shock in the three-equation New Keynesian model with the hybrid NKPC (6.6). Responses are in normalized model units; horizons are six-month periods. “Treated counterfactual” endows all price setters with a common signal of the precision calibrated in Section 6.2.

APPENDIX

A Descriptive Statistics

Figure A.1: Distribution Of 12-Months Ahead Inflation Forecasts

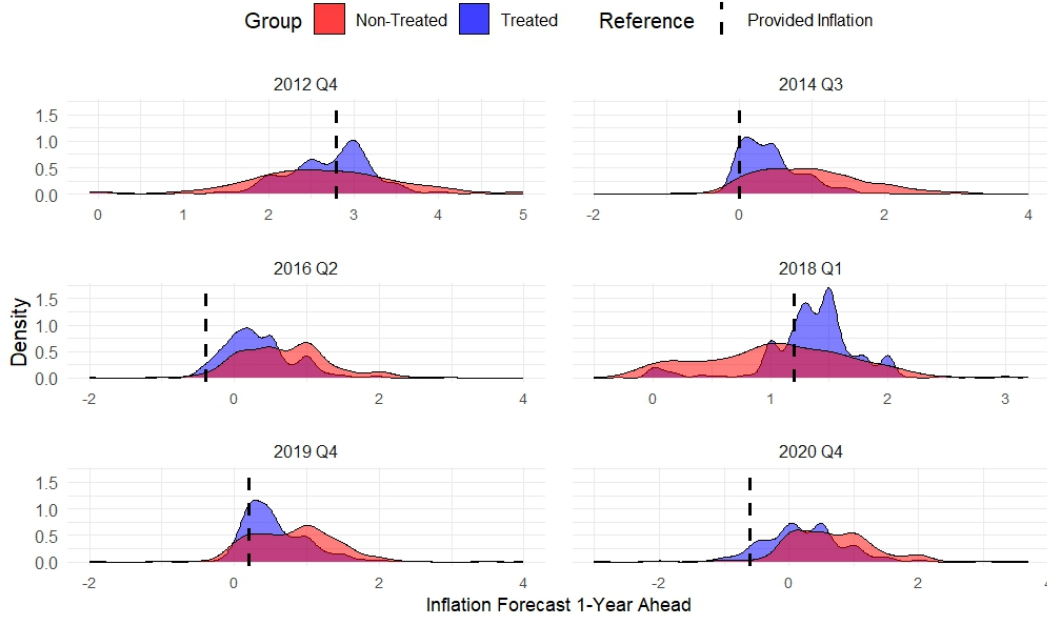


Figure A.1 plots kernel density estimates of 12-month-ahead inflation expectations for treated and non-treated firms. The vertical dashed line indicates the most recent 12-month inflation rate provided to treated firms at the time of the survey. Each panel corresponds to a different survey quarter.

B Additional Robustness Checks

B.1 Measurement Error

A standard concern in individual error-on-revision regressions is that measurement error in reported forecasts may mechanically induce a negative coefficient. If the information

treatment also reduces reporting error, this could make the treated coefficient appear less negative even without changing belief updating. We address this concern using cross-horizon revisions that break the direct overlap between the dependent variable and the revision.

The first exercise regresses the six-month-ahead forecast error on the one-year forecast revision, defined as the one-year-ahead forecast in quarter t minus the two-year-ahead forecast in quarter $t - 4$. This is the specification reported in Table B.1. The second exercise regresses the one-year-ahead forecast error on the six-month forecast revision, defined as the six-month-ahead forecast in quarter t minus the one-year-ahead forecast from two quarters earlier; Table B.2 reports these results. In both cases, the pattern mirrors the baseline: the non-treated coefficient is negative and significant, the treated coefficient is closer to zero and not statistically distinguishable from zero, and the treated–non-treated difference is positive but imprecisely estimated. These results do not support the interpretation that the baseline treatment effect is driven by the treatment mechanically reducing measurement-error bias.

Table B.1: Measurement-error robustness: six-month forecast error on one-year-to-two-year revision

Dependent Variable: Sample:	Forecast Error 6 (Months)					
	Treated	Non-treated	Pooled	Treated	Non-treated	Pooled
Forecast Revision (1 Year - 2 Year)	-0.06 (0.06)	-0.12** (0.05)		-0.09 (0.06)	-0.17*** (0.05)	
β_I^{NT}			-0.12** (0.05)			-0.17*** (0.05)
β_{DIFF}			0.06 (0.05)			0.08 (0.05)
<i>Fixed-effects</i>						
firm				Yes	Yes	
firm $\times$ treatment						Yes
<i>Fit statistics</i>						
Observations	13,204	4,847	18,051	13,204	4,847	18,051
R ²	0.00534	0.01269	0.00837	0.20275	0.31967	0.23941
Within R ²				0.01147	0.02684	0.01576

Addresses mechanical correlation by regressing the 6-month-ahead error on a cross-horizon revision (the 1-year-ahead minus the 2-year-ahead forecast made four quarters earlier). The treated–non-treated difference is not significant: $\beta_{DIFF} = 0.08$ ($p = 0.107$) with fixed effects and 0.06 ($p = 0.235$) without. Standard errors clustered by firm and quarter.

Table B.2: Measurement-error robustness: one-year forecast error on six-month-to-one-year revision

Dependent Variable: Sample:	Forecast Error 12 (Months)					
	Treated	Non-treated	Pooled	Treated	Non-treated	Pooled
Forecast Revision (6 Months - 1 Year)	-0.08 (0.13)	-0.14** (0.06)		-0.14 (0.11)	-0.27*** (0.05)	
β_I^{NT}			-0.14** (0.06)			-0.27*** (0.05)
β_{DIFF}			0.06 (0.10)			0.12 (0.09)
<i>Fixed-effects</i>						
firm				Yes	Yes	
firm $\times$ treatment						Yes
<i>Fit statistics</i>						
Observations	15,246	5,792	21,038	15,246	5,792	21,038
R ²	0.00193	0.00522	0.00286	0.24640	0.33571	0.27151
Within R ²				0.00697	0.02414	0.01142

Table B.2 reports regressions of the 1-year-ahead forecast error on the 6-month forecast revision (the 6-month-ahead forecast in quarter t minus the 1-year-ahead forecast from two quarters earlier), with and without firm fixed effects. Columns (1)–(3) exclude firm fixed effects; columns (4)–(6) include them. The Pooled columns (3) and (6) report the difference specification (Equation (3.4)). The difference is not significant: $\beta_{DIFF} = 0.12$ ($p = 0.164$) with fixed effects and 0.06 ($p = 0.580$) without. Standard errors are clustered by firm and quarter. Significance levels: *** $\Rightarrow p < 0.01$, ** $\Rightarrow p < 0.05$, * $\Rightarrow p < 0.1$.

B.2 Pre-COVID Sample Robustness

To ensure that our findings are not driven by the unusual macroeconomic environment during the COVID-19 period, we restrict the sample to the years 2010-2018. The results continue to hold, confirming that they are not specific to pandemic-era dynamics. Restricting to the pre-COVID period, the pooled difference remains positive but is not statistically distinguishable from zero ($\beta_{DIFF} = 0.10$, $p = 0.20$).

Table B.3: β_I Estimates: Pre-COVID Sample

Dependent Variable: Sample:	Individual Forecast Error (6 Months)					
	Treated	Non-treated	Pooled	Treated	Non-treated	Pooled
β_I	-0.15* (0.08)	-0.28*** (0.06)		-0.15* (0.08)	-0.25*** (0.07)	
β_I^{NT}			-0.28*** (0.06)			-0.25*** (0.07)
β_{DIFF}			0.14** (0.06)			0.10 (0.08)
<i>Fixed-effects</i>						
firm	Yes	Yes		Yes	Yes	
firm $\times$ treatment			Yes			Yes
<i>Fit statistics</i>						
Observations	15,266	5,805	21,071	11,961	4,423	16,384
R ²	0.18723	0.32974	0.23589	0.21586	0.39279	0.27698
Within R ²	0.01734	0.05621	0.02872	0.01957	0.04752	0.02748

Table B.3 reports estimates of Equation (3.2) for the full sample (columns (1)–(3)) and a restricted pre-COVID sample ending in 2019 (columns (4)–(6)), for treated and non-treated firms; all regressions include firm fixed effects. The Pooled columns (3) and (6) report the difference specification (Equation (3.4)). The difference is $\beta_{DIFF} = 0.14$ ($p = 0.040$; wild-cluster bootstrap $p = 0.044$) full-sample and 0.10 ($p = 0.204$) in the pre-COVID window. Standard errors are clustered at the firm and quarter level. This table assesses whether our main results are sensitive to the inclusion of the COVID-19 period. Significance levels: *** $\Rightarrow p < 0.01$, ** $\Rightarrow p < 0.05$, * $\Rightarrow p < 0.1$.

B.3 Drop Inattentive Firms

To ensure that our findings are not driven by treated firms reporting as their inflation forecast the same value that was given to them as treatment, we conduct another robustness check, where for the treated firms, we drop firm-quarter observations in the treated subsample where both the 6-month and 1-year ahead inflation forecasts are exactly equal to the inflation release that was provided to firms as part of the treatment.

Table B.4: β_I Regressions: Drop inattentive firms

Dependent Variable: Sample:	Individual Forecast Error (6 Months)			
	Treat	Non-Treat	Treat-Drop	Pooled
β_I	-0.15*	-0.28***	-0.14*	
	(0.08)	(0.06)	(0.08)	
β_I^{NT}				-0.28***
				(0.06)
β_{DIFF} (Drop – NT)				0.14**
				(0.06)
<i>Fixed-effects</i>				
firm	Yes	Yes	Yes	
firm $\times$ treatment				Yes
<i>Fit statistics</i>				
Observations	15,266	5,805	14,382	20,187
R ²	0.18723	0.32974	0.19303	0.24062
Within R ²	0.01734	0.05621	0.01590	0.02809

Table B.4 reports estimates of Equation (3.2) after excluding inattentive firms – those in the treated sample reporting 6-month and 1-year-ahead forecasts exactly equal to the inflation release provided as treatment. Columns (1)–(3) report the treated, non-treated, and inattentive-dropped treated samples; all include firm fixed effects. The Pooled column (4) reports the difference specification (Equation (3.4)) comparing the inattentive-dropped treated sample to non-treated firms: $\beta_{DIFF} = 0.14$ ($p = 0.032$; wild-cluster bootstrap $p = 0.033$). Standard errors are clustered by firm and quarter. This robustness check ensures that our results are not driven by firms that mechanically repeat past forecasts rather than responding to new information. Significance levels: *** $\Rightarrow p < 0.01$, ** $\Rightarrow p < 0.05$, * $\Rightarrow p < 0.1$.

B.4 Heterogeneity Analysis by Size

We conduct another robustness check to our individual level results where (3.2) is estimated separately on a sample of relatively small and large firms. Small (large) firms in any given quarter are defined as having fewer (greater) than 300 employees in that particular quarter. We estimate this specification with and without fixed effects. Tables B.5 and B.6 report the estimates. The pooled difference is significant for large firms ($\beta_{DIFF} = 0.19$, $p = 0.01$) but not for small firms (0.10 , $p = 0.16$). The point estimate stays positive in both; the loss of significance among small firms reflects the smaller cell, not a change in sign, since splitting the panel by size sharply cuts the treated observations in each group.

Table B.5: β_I Estimates: Small Firms

Dependent Variable: Sample:	Individual Forecast Error (6 Months)					
	Treated	Non-treated	Pooled	Treated	Non-treated	Pooled
β_I	-0.14 (0.09)	-0.21*** (0.06)		-0.19** (0.08)	-0.29*** (0.05)	
β_I^{NT}			-0.21*** (0.06)			-0.29*** (0.05)
β_{DIFF}			0.07 (0.07)			0.10 (0.07)
<i>Fixed-effects</i>						
firm				Yes	Yes	
firm $\times$ treatment						Yes
<i>Fit statistics</i>						
Observations	9,502	3,802	13,304	9,502	3,802	13,304
R ²	0.01506	0.02537	0.02153	0.22215	0.35438	0.26937
Within R ²				0.03098	0.06305	0.04083

Table B.5 reports estimates of equation (3.2), with and without firm fixed effects, for treated and non-treated firms with fewer than 300 employees. Columns (1)–(3) exclude firm fixed effects; columns (4)–(6) include them. The Pooled columns (3) and (6) report the difference specification (Equation (3.4)). The difference is not significant: $\beta_{DIFF} = 0.10$ ($p = 0.164$) with fixed effects and 0.07 ($p = 0.365$) without. All specifications use clustered standard errors at the firm and quarter level. Significance levels: *** $\Rightarrow p < 0.01$, ** $\Rightarrow p < 0.05$, * $\Rightarrow p < 0.1$.

Table B.6: β_I Estimates: Large Firms

Dependent Variable: Sample:	Individual Forecast Error (6 Months)					
	Treated	Non-treated	Pooled	Treated	Non-treated	Pooled
β_I	-0.02 (0.09)	-0.19*** (0.07)		-0.07 (0.08)	-0.26*** (0.06)	
β_I^{NT}			-0.19*** (0.07)			-0.26*** (0.06)
β_{DIFF}			0.17* (0.08)			0.19** (0.07)
<i>Fixed-effects</i>						
firm				Yes	Yes	
firm $\times$ treatment						Yes
<i>Fit statistics</i>						
Observations	5,487	1,901	7,388	5,487	1,901	7,388
R ²	0.00044	0.01903	0.00916	0.16483	0.31884	0.21289
Within R ²				0.00425	0.04343	0.01454

Table B.6 reports estimates of equation (3.2), with and without firm fixed effects, for treated and non-treated firms with more than 300 employees. Columns (1)–(3) exclude firm fixed effects; columns (4)–(6) include them. The Pooled columns (3) and (6) report the difference specification (Equation (3.4)). The difference is $\beta_{DIFF} = 0.19$ ($p = 0.010$) with fixed effects and 0.17 ($p = 0.051$) without. All specifications use clustered standard errors at the firm and quarter level. Significance levels: *** $\Rightarrow p < 0.01$, ** $\Rightarrow p < 0.05$, * $\Rightarrow p < 0.1$.

B.5 Heterogeneity by Financial Constraint and Demand Sensitivity

As a further robustness check, we estimate (3.2) separately on firms that differ in their exposure to financial and demand conditions. In each quarter we classify a firm as financially constrained if it reports credit conditions to be a strong constraint on its activity, and as demand-sensitive if it reports its business situation to be strongly affected by demand; the complementary groups are financially unconstrained and demand-insensitive, respectively. We estimate the specification with and without fixed effects within each group. The results are reported in Tables B.7–B.10. The point estimate of the treatment-induced difference β_{DIFF} remains positive and of similar magnitude to the full sample across all four subsamples, though—as with the size cuts—slicing the sample reduces power, so the difference is estimated less precisely within any single group.

Table B.7: β_I Estimates: Financially Constrained Firms

Dependent Variable: Sample:	Individual Forecast Error (6 Months)					
	Treated	Non-treated	Pooled	Treated	Non-treated	Pooled
β_I	-0.07 (0.08)	-0.17** (0.08)		-0.16*** (0.05)	-0.27*** (0.09)	
β_I^{NT}			-0.17** (0.08)			-0.27*** (0.08)
β_{DIFF}			0.10 (0.11)			0.11 (0.10)
<i>Fixed-effects</i>						
firm				Yes	Yes	
firm $\times$ treatment						Yes
<i>Fit statistics</i>						
Observations	1,269	396	1,665	1,269	396	1,665
R ²	0.00445	0.01805	0.00938	0.46827	0.63206	0.51161
Within R ²				0.02375	0.06895	0.03295

Table B.7 reports estimates of equation (3.2), with and without firm fixed effects, for treated and non-treated financially constrained firms (those reporting credit conditions to be a strong constraint on activity). Columns (1)–(3) exclude firm fixed effects; columns (4)–(6) include them. The Pooled columns (3) and (6) report the difference specification (Equation (3.4)). The difference is not statistically significant – $\beta_{DIFF} = 0.11$ ($p = 0.270$; wild-cluster bootstrap $p = 0.254$) with fixed effects and 0.10 ($p = 0.346$) without – reflecting the small subsample. Standard errors are clustered by firm and quarter. Significance levels: *** $\Rightarrow p < 0.01$, ** $\Rightarrow p < 0.05$, * $\Rightarrow p < 0.1$.

Table B.8: β_I Estimates: Financially Unconstrained Firms

Dependent Variable: Sample:	Individual Forecast Error (6 Months)					
	Treated	Non-treated	Pooled	Treated	Non-treated	Pooled
β_I	-0.12 (0.09)	-0.21*** (0.06)		-0.19** (0.08)	-0.30*** (0.06)	
β_I^{NT}			-0.21*** (0.06)			-0.30*** (0.06)
β_{DIFF}			0.09 (0.08)			0.11 (0.07)
<i>Fixed-effects</i>						
firm				Yes	Yes	
firm $\times$ treatment						Yes
<i>Fit statistics</i>						
Observations	9,944	3,963	13,907	9,944	3,963	13,907
R ²	0.01010	0.02449	0.01737	0.24054	0.33471	0.27396
Within R ²				0.02700	0.05754	0.03658

Table B.8 reports estimates of equation (3.2), with and without firm fixed effects, for treated and non-treated financially unconstrained firms. Columns (1)–(3) exclude firm fixed effects; columns (4)–(6) include them. The Pooled columns (3) and (6) report the difference specification (Equation (3.4)): $\beta_{DIFF} = 0.11$ ($p = 0.102$; wild-cluster bootstrap $p = 0.112$) with fixed effects and 0.09 ($p = 0.236$) without. Standard errors are clustered by firm and quarter. Significance levels: *** $\Rightarrow p < 0.01$, ** $\Rightarrow p < 0.05$, * $\Rightarrow p < 0.1$.

Table B.9: β_I Estimates: Demand-Sensitive Firms

Dependent Variable: Sample:	Individual Forecast Error (6 Months)					
	Treated	Non-treated	Pooled	Treated	Non-treated	Pooled
β_I	-0.10 (0.10)	-0.21*** (0.07)		-0.20** (0.07)	-0.33*** (0.07)	
β_I^{NT}			-0.21*** (0.07)			-0.33*** (0.06)
β_{DIFF}			0.10 (0.10)			0.13* (0.07)
<i>Fixed-effects</i>						
firm				Yes	Yes	
firm $\times$ treatment						Yes
<i>Fit statistics</i>						
Observations	3,316	1,241	4,557	3,316	1,241	4,557
R ²	0.00724	0.02331	0.01841	0.38265	0.47619	0.41517
Within R ²				0.02950	0.07477	0.04235

Table B.9 reports estimates of equation (3.2), with and without firm fixed effects, for treated and non-treated demand-sensitive firms (those reporting business conditions to be strongly affected by demand). Columns (1)–(3) exclude firm fixed effects; columns (4)–(6) include them. The Pooled columns (3) and (6) report the difference specification (Equation (3.4)): $\beta_{DIFF} = 0.13$ ($p = 0.071$; wild-cluster bootstrap $p = 0.066$) with fixed effects and 0.10 ($p = 0.314$) without. Standard errors are clustered by firm and quarter. Significance levels: *** $\Rightarrow p < 0.01$, ** $\Rightarrow p < 0.05$, * $\Rightarrow p < 0.1$.

Table B.10: β_I Estimates: Demand-Insensitive Firms

Dependent Variable: Sample:	Individual Forecast Error (6 Months)					
	Treated	Non-treated	Pooled	Treated	Non-treated	Pooled
β_I	-0.05 (0.09)	-0.18*** (0.06)		-0.13* (0.07)	-0.27*** (0.06)	
β_I^{NT}			-0.18*** (0.06)			-0.27*** (0.06)
β_{DIFF}			0.13 (0.08)			0.14* (0.07)
<i>Fixed-effects</i>						
firm				Yes	Yes	
firm $\times$ treatment						Yes
<i>Fit statistics</i>						
Observations	5,638	2,205	7,843	5,638	2,205	7,843
R ²	0.00204	0.01696	0.00998	0.28339	0.38004	0.31723
Within R ²				0.01457	0.04868	0.02497

Table B.10 reports estimates of equation (3.2), with and without firm fixed effects, for treated and non-treated demand-insensitive firms. Columns (1)–(3) exclude firm fixed effects; columns (4)–(6) include them. The Pooled columns (3) and (6) report the difference specification (Equation (3.4)): $\beta_{DIFF} = 0.14$ ($p = 0.054$; wild-cluster bootstrap $p = 0.050$) with fixed effects and 0.13 ($p = 0.129$) without. Standard errors are clustered by firm and quarter. Significance levels: *** $\Rightarrow p < 0.01$, ** $\Rightarrow p < 0.05$, * $\Rightarrow p < 0.1$.

B.6 Fixed Effects and Short-Panel Concerns

Our baseline individual-level specification estimates equation (3.2) with firm fixed effects, so that β_I is identified from within-forecaster variation in revisions and time-invariant heterogeneity in forecasting behavior is absorbed. Because our panel is relatively short—and the non-treated series begins only in 2012 Q3—fixed-effects estimators may be subject to Nickell bias. As a robustness check, Table B.11 reports the individual-level estimates with and without firm fixed effects. The non-treated coefficient remains negative and statistically significant, while the treated coefficient is less negative and statistically insignificant; the pooled treated–non-treated difference is correspondingly smaller and no longer significant ($\beta_{DIFF} = 0.11$, $p = 0.14$), with the treated coefficient itself indistinguishable from zero. These results are consistent with our fixed-effects baseline.

Table B.11: Individual-level regressions with and without firm fixed effects

Dependent Variable: Sample:	Individual Forecast Error (6 Months)					
	Treated	Non-treated	Pooled	Treated	Non-treated	Pooled
β_I	-0.10 (0.09)	-0.20*** (0.06)		-0.15* (0.08)	-0.28*** (0.06)	
β_I^{NT}			-0.20*** (0.06)			-0.28*** (0.06)
β_{DIFF}			0.11 (0.07)			0.14** (0.06)
<i>Fixed-effects</i>						
firm				Yes	Yes	
firm $\times$ treatment						Yes
<i>Fit statistics</i>						
Observations	15,266	5,805	21,071	15,266	5,805	21,071
R ²	0.00710	0.02380	0.01549	0.18723	0.32974	0.23589
Within R ²				0.01734	0.05621	0.02872

Estimates of equation (3.2) for treated and non-treated firms, with (columns 4–6) and without (columns 1–3) firm fixed effects; the Pooled columns report the difference specification (3.4). The treated–non-treated difference is $\beta_{DIFF} = 0.14$ ($p = 0.040$; bootstrap $p = 0.044$) with fixed effects and 0.11 ($p = 0.140$) without. Standard errors clustered by firm and quarter.

C Proofs

This section of the appendix presents detailed derivations of the consensus and individual-level error-on-revision regression coefficients and the analytical formula for the cross sectional variance of forecasts under the assumptions detailed in the theory section. Throughout this section, we maintain the following assumptions about the data-generating process and the signal structure.

Specifically, $x_t = \rho x_{t-1} + u_t$; $u_t \sim N(0, \sigma_u^2)$. Forecasters observe an idiosyncratic noisy signal $s_t^i = x_t + \epsilon_{it}$; $\epsilon_{it} \sim N(0, \sigma_\epsilon^2)$. We model the survey's information intervention as giving treated firms access to a public signal about recent inflation at the time forecasts are formed. This public signal is denoted by: $s_t = x_t + \nu_t$; $\nu_t \sim N(0, \sigma_\nu^2)$. The non-treated case corresponds to shutting down this public signal (equivalently, letting its variance go to infinity, $\sigma_\nu^2 \rightarrow \infty$, i.e., its precision goes to zero.)²³.

C.1 Noisy Memory

Here, we assume that forecasters face constraints in their ability to recall past information. Limited ability to recall past information causes forecasters to place too much weight on current news, causing individual-level overreaction. This section adapts the model in Section 2 of Sung (2026), replacing the assumption of endogenous processing of external news with exogenous noisy signals, as in equations (2) and (3). Let m_{it} denote the agent's memory state at the start of period t , while $M_{it} = (m_{it-1}, s_{it-1}, s_{t-1})$ denotes the prior knowledge that is available to the agent before she observes the noisy signals at time t . Following Sung (2026), we assume that the mental representation of existing knowledge takes the following form:

$$m_{it} = \Lambda_t M_{it} + \tilde{\omega}_{it}; \quad \tilde{\omega}_{it} \sim N(0, \Sigma_{\omega t}) \quad (\text{C.1})$$

The agent also faces a constraint on the mutual information between m_{it} and M_{it} :

$$\mathcal{I}(m_{it}, M_{it}) \leq -\frac{\log \phi_m}{2} \quad (\text{C.2})$$

²³This public-signal representation is a simple way to model the fact that the survey makes recent inflation information salient at the time of forecasting. In a costly information-processing or rational-inattention interpretation, the intervention can be viewed as lowering the cost of acquiring/processing publicly available inflation data.

In this case, the optimal memory representation takes the following form:

$$m_{it} = \lambda_t \mathbb{E}[x_t | M_{it}] + \omega_{it}; \quad \omega_{it} \sim N(0, \sigma_{\omega_t}) \quad (\text{C.3})$$

where $\lambda_t \in [0, 1]$. Consistent with our earlier notation, $\mathbb{E}[x_t | M_{it}] = x_{t|t-1}^i$. In order to ensure that σ_{ω_t} is fully determined by λ_t , we assume the following normalization:

$$\text{Cov}[\mathbb{E}[x_t | M_{it} - m_{it}], m_{it}] = 0 \quad (\text{C.4})$$

which implies that $\sigma_{\omega_t}^2 = \lambda_t(1 - \lambda_t) \text{Var}(\mathbb{E}[x_t | M_{it}])$. It can then be shown that:

$$\mathcal{I}(m_{it}, M_{it}) \leq -\frac{\log(1 - \lambda_t)}{2} \Rightarrow \lambda_t = \lambda = 1 - \phi_m \quad (\text{C.5})$$

Following Proposition 2 of Sung (2026), the prior distribution of x_t conditional on memory state m_{it} (before observing noisy signals)

$$x_t | m_{it} \sim N(x_{t|t}^{i,m}, \Sigma_{t|t}^{i,m}) \quad (\text{C.6})$$

$$x_{t|t}^{i,m} = \lambda x_{t|t-1}^i + \omega_{it}; \quad (\text{C.7})$$

$$\Sigma_{t|t}^{i,m} = (1 - \lambda) \frac{\sigma_u^2}{1 - \rho^2} + \lambda \Sigma_{t|t-1}^i \quad (\text{C.8})$$

Standard Kalman filtering with normal priors and normal shocks implies that the posterior distribution of x_t conditional on memory state m_{it} , private signal s_{it} and public signal s_t is:

$$x_t | (m_{it}, s_{it}, s_t) \sim N(x_{t|t}^i, \Sigma_{t|t}^i) \quad (\text{C.9})$$

$$x_{t|t}^i = x_{t|t}^{i,m} + K_{1t}(s_{it} - x_{t|t}^{i,m}) + K_{2t}(s_t - x_{t|t}^{i,m}) \quad (\text{C.10})$$

$$\Sigma_{t|t}^i = (1 - K_t) \Sigma_{t|t}^{i,m} \quad (\text{C.11})$$

$$K_{1t} = \frac{\Sigma_{t|t}^{i,m} \sigma_\nu^2}{\Sigma_{t|t}^{i,m} (\sigma_\nu^2 + \sigma_\epsilon^2)} \quad (\text{C.12})$$

$$K_{2t} = \frac{\Sigma_{t|t}^{i,m} \sigma_\epsilon^2}{\Sigma_{t|t}^{i,m} (\sigma_\nu^2 + \sigma_\epsilon^2)} \quad (\text{C.13})$$

where K_{1t} is the Kalman gain loading on the private signal, K_{2t} is the Kalman gain on the public signal and $K_t = K_{1t} + K_{2t}$. The usual Riccatti equation is given by:

$$\Sigma_{t+1|t}^i = \rho^2 \Sigma_{t|t}^i + \sigma_u^2 = \rho^2(1 - K_t) \Sigma_{t|t}^{i,m} + \sigma_u^2 \quad (\text{C.14})$$

We solve for the steady-state variance by substituting the expression for expression for $\Sigma_{t|t}^{i,m}$ from equation (C.8) into the above equation and imposing $\Sigma_{t+1|t}^i = \Sigma_{t|t-1}^i = \Sigma$. The steady-state variance is then given by the following expression:

$$\Sigma = \frac{-B + \sqrt{B^2 - 4AC}}{2A}, \quad (\text{C.15})$$

$$A = \lambda (\sigma_\epsilon^2 + \sigma_\nu^2), \quad (\text{C.16})$$

$$B = (\sigma_\epsilon^2 + \sigma_\nu^2) [(1 - \lambda)\sigma_u^2 - \lambda\sigma_x^2] + \sigma_\epsilon^2 \sigma_\nu^2 [1 - \lambda\rho^2], \quad (\text{C.17})$$

$$C = -[(1 - \lambda)\sigma_u^2 \sigma_x^2 (\sigma_\epsilon^2 + \sigma_\nu^2) + \sigma_\epsilon^2 \sigma_\nu^2 (\sigma_u^2 + (1 - \lambda)\sigma_x^2)], \quad (\text{C.18})$$

where $\sigma_x^2 = \frac{\sigma_u^2}{1 - \rho^2}$. Plugging the steady state variance $\Sigma_{t|t-1}^i = \Sigma$ into the expression for $\Sigma_{t|t}^{i,m}$ and subsequently into K_{1t} and K_{2t} allows us to derive the steady-state Kalman gains K_1 and K_2 . Given the steady state Kalman gains, we can write $F_{i,t}(x_t) = x_{t|t}^i$ using the standard Kalman updating equation:

$$x_{t|t}^i = (1 - K)\lambda x_{t|t-1}^i + Kx_t + K_1\epsilon_{it} + K_2\nu_t + (1 - K)\omega_{it} \quad (\text{C.19})$$

$$= [(1 - K)\lambda + K]x_{t|t-1}^i + K(x_t - x_{t|t-1}^i) + K_1\epsilon_{it} + K_2\nu_t + (1 - K)\omega_{it}. \quad (\text{C.20})$$

The variance of the forecast revision can be expressed as follows:

$$\text{Var}(x_{t|t}^i - x_{t|t-1}^i) = \text{Var}(x_{t|t}^i) + \text{Var}(x_{t|t-1}^i) - 2 \text{Cov}(x_{t|t}^i, x_{t|t-1}^i) \quad (\text{C.21})$$

Using the expression for $x_{t|t}^i$:

$$\text{Cov}(x_{t|t}^i, x_{t|t-1}^i) = [(1 - K)\lambda + K] \text{Var}(x_{t|t-1}^i). \quad (\text{C.22})$$

Using $\text{Var}(x_{t+1|t}^i) = \text{Var}(x_{t|t-1}^i) = \rho^2 \text{Var}(x_{t|t}^i)$, it then follows that:

$$\text{Var}(x_{t|t}^i - x_{t-1|t-1}^i) = \text{Var}(x_{t|t-1}^i) [\rho^{-2} + 1 - 2((1-K)\lambda + K)]. \quad (\text{C.23})$$

Next, note that contemporaneous forecast error is given by:

$$x_t - x_{t|t}^i = (1-K)x_t - (1-K)\lambda x_{t|t-1}^i - K_1\epsilon_{it} - K_2\nu_t - (1-K)\omega_{it}. \quad (\text{C.24})$$

Taking the covariance between the forecast error $(x_t - x_{t|t}^i)$ and the forecast revision $(x_{t|t}^i - x_{t|t-1}^i)$ and using the expression for $x_t - x_{t|t}^i$ gives:

$$\text{Cov}(x_t - x_{t|t}^i, x_{t|t}^i - x_{t|t-1}^i) = -\text{Cov}(x_t - x_{t|t}^i, x_{t|t-1}^i) \quad (\text{C.25})$$

$$= -(1-K)(1-\lambda) \text{Var}(x_{t|t-1}^i). \quad (\text{C.26})$$

Combining the above results, the individual-level error-on-revision coefficient is:

$$\beta_I = \frac{\text{Cov}(x_{t+h} - x_{t+h|t}^i, x_{t+h|t}^i - x_{t+h|t-1}^i)}{\text{Var}(x_{t+h|t}^i - x_{t+h|t-1}^i)} \quad (\text{C.27})$$

$$= \frac{\text{Cov}(x_t - x_{t|t}^i, x_{t|t}^i - x_{t-1|t-1}^i)}{\text{Var}(x_{t|t}^i - x_{t-1|t-1}^i)} \quad (\text{C.28})$$

$$= -\frac{(1-K)(1-\lambda)}{\rho^{-2} + 1 - 2[(1-K)\lambda + K]}. \quad (\text{C.29})$$

Integrating (C.24) across forecasters i , we can write down the consensus forecast error:

$$x_t - x_{t|t} = (1-K)(x_t - x_{t|t-1}) + (1-\lambda)(1-K)x_{t|t-1} - K_2\nu_t \quad (\text{C.30})$$

Subtracting $x_{t|t-1}^i$ from equation (C.19) and integrating over i , we can derive the consensus forecast revision:

$$x_{t|t} - x_{t|t-1} = (1 - K)(\lambda - 1)x_{t|t-1} + K(x_t - x_{t|t-1}) + K_2 \nu_t \quad (\text{C.31})$$

$$\Rightarrow \frac{x_{t|t} - x_{t|t-1}}{K} = \frac{(1 - K)(\lambda - 1)x_{t|t-1}}{K} + (x_t - x_{t|t-1}) + \frac{K_2 \nu_t}{K} \quad (\text{C.32})$$

$$\Rightarrow (x_{t|t} - x_{t|t-1}) \left(\frac{1 - K}{K} \right) = x_t - x_{t|t} - \frac{(1 - \lambda)(1 - K)}{K} x_{t|t-1} + \frac{K_2}{K} \nu_t \quad (\text{C.33})$$

$$\Rightarrow x_t - x_{t|t} = \frac{1 - K}{K} (x_{t|t} - x_{t|t-1} + (1 - \lambda)x_{t|t-1}) - \frac{K_2}{K} \nu_t \quad (\text{C.34})$$

where the third line follows from adding and subtracting $x_{t|t}$ on the RHS and rearranging. Then, β_C is given by the following expression:

$$\beta_C = \frac{\text{Cov}(x_{t+h} - x_{t+h|t}, x_{t+h|t} - x_{t+h|t-1})}{\text{Var}(x_{t+h|t} - x_{t+h|t-1})} \quad (\text{C.35})$$

$$= \frac{\text{Cov}(x_t - x_{t|t}, x_{t|t} - x_{t|t-1})}{\text{Var}(x_{t|t} - x_{t|t-1})} \quad (\text{C.36})$$

$$= \frac{\frac{1-K}{K} \text{Var}(x_{t|t} - x_{t|t-1}) + \frac{1-K}{K} (1 - \lambda) \text{Cov}(x_{t|t} - x_{t|t-1}, x_{t|t-1}) - \frac{K_2^2}{K} \sigma_\nu^2}{\text{Var}(x_{t|t} - x_{t|t-1})} \quad (\text{C.37})$$

$$= \frac{1 - K}{K} + \frac{1 - K}{K} (1 - \lambda) \frac{\text{Cov}(x_{t|t} - x_{t|t-1}, x_{t|t-1})}{\text{Var}(x_{t|t} - x_{t|t-1})} - \frac{K_2^2}{K} \frac{\sigma_\nu^2}{\text{Var}(x_{t|t} - x_{t|t-1})} \quad (\text{C.38})$$

which is the expression given in equation (5.10), with $\text{Cov}(x_{t|t} - x_{t|t-1}, x_{t|t-1}) = C_{x_{t|t-1}, agg}$ and $\text{Var}(x_{t|t} - x_{t|t-1}) = V_{agg}$ (also used is the fact that $\text{Cov}(\nu_t, x_{t|t} - x_{t|t-1}) = K_2 \sigma_\nu^2$). Furthermore, using equation (C.31) we can write:

$$\text{Cov}(x_{t|t} - x_{t|t-1}, x_{t|t-1}) \quad (\text{C.39})$$

$$= \text{Cov}((1 - K)(\lambda - 1)x_{t|t-1} + K(x_t - x_{t|t-1}) + K_2 \nu_t, x_{t|t-1}) \quad (\text{C.40})$$

$$= K \text{Cov}(x_t - x_{t|t-1}, x_{t|t-1}) + (\lambda - 1)(1 - K) \text{Var}(x_{t|t-1}) \quad (\text{C.41})$$

Again, using equation (C.31), we can write:

$$\text{Var}(x_{t|t} - x_{t|t-1}) \quad (\text{C.42})$$

$$= K^2 \text{Var}(x_t - x_{t|t-1}) + (1 - K)^2 (\lambda - 1)^2 \text{Var}(x_{t|t-1}) + K_2^2 \sigma_\nu^2 + 2 \text{Cov}(x_t - x_{t|t-1}, x_{t|t-1}) \quad (\text{C.43})$$

The three expressions we need to obtain expressions to calculate β_C are: $\text{Var}(x_t - x_{t|t-1})$, $\text{Cov}(x_t - x_{t|t-1}, x_{t|t-1})$ and $\text{Var}(x_{t|t-1})$. Let $x_t - x_{t|t-1} = a_t$ and $x_{t|t-1} = b_t$. It can then be shown that:

$$a_{t+1} = \rho(1 - K)a_t + \rho(1 - K)(1 - \lambda)b_t + u_{t+1} - \rho K_2 \nu_t \quad (\text{C.44})$$

$$b_{t+1} = \rho K a_t + \rho(K + \lambda(1 - K))b_t + \rho K_2 \nu_t \quad (\text{C.45})$$

Writing this system as a VAR:

$$\begin{bmatrix} a_{t+1} \\ b_{t+1} \end{bmatrix} = \begin{bmatrix} \rho(1 - K) & \rho(1 - K)(1 - \lambda) \\ \rho K & \rho[K + \lambda(1 - K)] \end{bmatrix} \begin{bmatrix} a_t \\ b_t \end{bmatrix} + \begin{bmatrix} 1 & -\rho K_2 \\ 0 & \rho K_2 \end{bmatrix} \begin{bmatrix} u_{t+1} \\ \nu_t \end{bmatrix} \quad (\text{C.46})$$

Solving for the steady-state variance of this 2x2 system allows us to obtain the expressions needed to calculate β_C , namely, $\text{Cov}(a_t, b_t)$, $\text{Var}(a_t)$ and $\text{Var}(b_t)$ ²⁴. Finally, we also show how to calculate the forecast dispersion. Integrating equation (C.19) across i and subtracting it from $x_{t|t}$, we have:

$$x_{t|t}^i - x_{t|t} = (1 - K)\lambda(x_{t|t-1}^i - x_{t|t-1}) + K_1 \epsilon_{it} + (1 - K)\omega_{it} \quad (\text{C.47})$$

$$= (1 - K)\lambda \rho (x_{t-1|t-1}^i - x_{t-1|t-1}) + K_1 \epsilon_{it} + (1 - K)\omega_{it} \quad (\text{C.48})$$

$$\text{Var}(x_{t|t}^i - x_{t|t}) = \frac{K_1^2 \sigma_\epsilon^2 + (1 - K)^2 \sigma_\omega^2}{1 - (1 - K)^2 \lambda^2 \rho^2} \quad (\text{C.49})$$

$$\Rightarrow \text{Var}(x_{t+h|t}^i - x_{t+h|t}) = \text{Var}(\rho^h (x_{t|t}^i - x_{t|t})) \quad (\text{C.50})$$

$$= \rho^{2h} \text{Var}(x_{t|t}^i - x_{t|t}) \quad (\text{C.51})$$

²⁴We call a matlab solver to solve this 2x2 system

C.2 Overconfidence

We now derive the consensus and individual error-on-revision regression coefficient in the presence of idiosyncratic private signals and a public signal, when forecasters perceive the variance of their idiosyncratic signal $\hat{\sigma}_\epsilon^2$ to be lower than its true variance σ_ϵ^2 . Given equations (5.1), (5.2) and (5.3), we can write the measurement and state equations of the system as:

$$y_t^i = Hx_t + w_t, \quad (\text{C.52})$$

$$x_t = \rho x_{t-1} + u_t \quad (\text{C.53})$$

where $y_t^i = [s_t^i, s_t]^i$, $H = [1, \quad 1]^i$, $w_t = [\epsilon_{it}, \quad \nu_t]^i$. Let $\Sigma_{t|t-1}^i = \mathbb{E} \left[\left(x_t - x_{t|t-1}^i \right)^2 \right]$ be their prior variance belief and $K_t^v = [K_{1t}, K_{2t}]^i$ denote the vector of Kalman gains. The Kalman recursion satisfies the following equations:

$$K_t^v = \Sigma_{t|t-1}^i H' (H \Sigma_{t|t-1}^i H' + R_{\text{perceived}})^{-1}, \quad (\text{C.54})$$

$$\Sigma_{t|t}^i = (1 - K_t^v H) \Sigma_{t|t-1}^i, \quad (\text{C.55})$$

$$\Sigma_{t+1|t}^i = \rho^2 \Sigma_{t|t}^i + \sigma_u^2. \quad (\text{C.56})$$

with the matrix $R_{\text{perceived}}$ given by the following expression:

$$\begin{bmatrix} \hat{\sigma}_\epsilon^2 & 0 \\ 0 & \sigma_\nu^2 \end{bmatrix} \quad (\text{C.57})$$

Using equations from the Kalman recursion, the steady-state prior variance $\Sigma_{t|t-1}^i = \Sigma$ is given by the following expression:

$$\Sigma = \frac{\sigma_u^2 (\hat{\sigma}_\epsilon^2 + \sigma_\nu^2) - (1 - \rho^2) \sigma_\nu^2 \hat{\sigma}_\epsilon^2 + \sqrt{[\sigma_u^2 (\hat{\sigma}_\epsilon^2 + \sigma_\nu^2) - (1 - \rho^2) \hat{\sigma}_\epsilon^2 \sigma_\nu^2]^2 + 4 (\hat{\sigma}_\epsilon^2 + \sigma_\nu^2) \sigma_\nu^2 \sigma_u^2 \hat{\sigma}_\epsilon^2}}{2 (\hat{\sigma}_\epsilon^2 + \sigma_\nu^2)} \quad (\text{C.58})$$

The steady state (distorted) Kalman gains on the public and private signals are then given by the following expressions:

$$K_1 = \frac{\Sigma \sigma_\nu^2}{(\hat{\sigma}_\epsilon^2 + \sigma_\nu^2) \Sigma + \hat{\sigma}_\epsilon^2 \sigma_\nu^2}, \quad (\text{C.59})$$

$$K_2 = \frac{\Sigma \hat{\sigma}_\epsilon^2}{(\hat{\sigma}_\epsilon^2 + \sigma_\nu^2) \Sigma + \hat{\sigma}_\epsilon^2 \sigma_\nu^2}. \quad (\text{C.60})$$

The Kalman updating formula at the steady-state Kalman gains gives us the following expression:

$$x_{t|t}^i = x_{t|t-1}^i + K_1 (s_t^i - x_{t|t-1}^i) + K_2 (s_t - x_{t|t-1}^i) \quad (\text{C.61})$$

$$\Rightarrow x_{t|t}^i - x_{t|t-1}^i = K(x_t - x_{t|t-1}^i) + K_1 \epsilon_{it} + K_2 \nu_t \quad (\text{C.62})$$

$$\Rightarrow \frac{x_{t|t}^i - x_{t|t-1}^i}{K} - \frac{K_1}{K} \epsilon_{it} - \frac{K_2}{K} \nu_t = x_t - x_{t|t-1}^i \quad (\text{C.63})$$

$$\Rightarrow x_t - x_{t|t}^i = (x_{t|t}^i - x_{t|t-1}^i) \left(\frac{1-K}{K} \right) - \frac{K_1}{K} \epsilon_{it} - \frac{K_2}{K} \nu_t \quad (\text{C.64})$$

We can now derive the expression for β_I using equation (C.64):

$$\beta_I = \frac{\text{Cov} \left(x_{t+h} - x_{t+h|t}^i, x_{t+h|t}^i - x_{t+h|t-1}^i \right)}{\text{Var} \left(x_{t+h|t}^i - x_{t+h|t-1}^i \right)} \quad (\text{C.65})$$

$$= \frac{\text{Cov} \left(x_t - x_{t|t}^i, x_{t|t}^i - x_{t-1|t-1}^i \right)}{\text{Var} \left(x_{t|t}^i - x_{t-1|t-1}^i \right)} \quad (\text{C.66})$$

$$= \frac{1-K}{K} - \frac{K_1^2 \sigma_\epsilon^2}{K V_{ind}} - \frac{K_2^2 \sigma_\nu^2}{K V_{ind}}. \quad (\text{C.67})$$

where $V_{ind} = \text{Var} \left(x_{t|t}^i - x_{t-1|t-1}^i \right)$. We have also used: $\text{Cov} \left(\epsilon_{it}, x_{t|t}^i - x_{t|t-1}^i \right) = K_1 \sigma_\epsilon^2$ and $\text{Cov} \left(\nu_t, x_{t|t}^i - x_{t|t-1}^i \right) = K_2 \sigma_\nu^2$. Finally, we can also show that:

$$x_{t+1} - x_{t+1|t}^i = \rho \left(x_t - x_{t|t}^i \right) + u_{t+1} \quad (\text{C.68})$$

$$= \rho \left[(1 - K) \left(x_t - x_{t|t-1}^i \right) - K_1 \epsilon_{it} - K_2 \nu_t \right] + u_{t+1} \quad (\text{C.69})$$

$$\Rightarrow \text{Var} \left(x_{t+1} - x_{t+1|t}^i \right) = \rho^2 (1 - K)^2 \text{Var} \left(x_t - x_{t|t-1}^i \right) + \rho^2 K_1^2 \sigma_\epsilon^2 + \rho^2 K_2^2 \sigma_\nu^2 + \sigma_u^2 \quad (\text{C.70})$$

$$\Rightarrow \text{Var} \left(x_t - x_{t|t-1}^i \right) = \frac{\rho^2 [K_1^2 \sigma_\epsilon^2 + K_2^2 \sigma_\nu^2]}{1 - \rho^2 (1 - K)^2} + \sigma_u^2 \quad (\text{C.71})$$

$$\text{Var} \left(x_{t|t}^i - x_{t|t-1}^i \right) = K^2 \text{Var} \left(x_t - x_{t|t-1}^i \right) + K_1^2 \sigma_\epsilon^2 + K_2^2 \sigma_\nu^2 \quad (\text{C.72})$$

$$= K^2 \left[\frac{\rho^2 (K_1^2 \sigma_\epsilon^2 + K_2^2 \sigma_\nu^2) + \sigma_u^2}{1 - \rho^2 (1 - K)^2} \right] + K_1^2 \sigma_\epsilon^2 + K_2^2 \sigma_\nu^2 \quad (\text{C.73})$$

$$= V_{ind} \quad (\text{C.74})$$

Integrating equation (C.64) over i and following very similar steps as above, we can write:

$$\beta_C = \frac{1 - K}{K} - \frac{K_2^2 \sigma_\nu^2}{K V_{agg}}. \quad (\text{C.75})$$

$$V_{agg} = \text{Var}(x_{t|t} - x_{t|t-1}) \quad (\text{C.76})$$

$$= K^2 \left[\frac{\rho^2 K_2^2 \sigma_\nu^2 + \sigma_u^2}{1 - \rho^2 (1 - K)^2} \right] + K_2^2 \sigma_\nu^2 \quad (\text{C.77})$$

Finally, we also show how to calculate the cross-sectional variance of forecasts. From equation (C.61), we integrate across i to get $x_{t|t}$. Subtracting $x_{t|t}$ from $x_{t|t}^i$, we have:

$$x_{t|t}^i - x_{t|t} = (1 - K)(x_{t|t-1}^i - x_{t|t-1}) + K_1 \epsilon_{it} \quad (\text{C.78})$$

$$= (1 - K) \rho \left(x_{t-1|t-1}^i - x_{t-1|t-1} \right) + K_1 \epsilon_{it} \quad (\text{C.79})$$

$$\Rightarrow \text{Var}(x_{t|t}^i - x_{t|t}) = \frac{K_1^2 \sigma_\epsilon^2}{1 - (1 - K)^2 \rho^2} \quad (\text{C.80})$$

$$\Rightarrow \text{Var}(x_{t+h|t}^i - x_{t+h|t}) = \text{Var} \left(\rho^h (x_{t|t}^i - x_{t|t}) \right) \quad (\text{C.81})$$

$$= \rho^{2h} \text{Var}(x_{t|t}^i - x_{t|t}) \quad (\text{C.82})$$

In our GMM exercises we target the ratio of the standard deviation of forecasts between treated and non-treated firms to discipline the value of σ_ν^2 .

C.3 Over-Extrapolation

We now derive the consensus and individual error-on-revision regression coefficients when agents misperceive the persistence of the process to be $\hat{\rho}$, where actual persistence is larger than perceived persistence. In this case, the measurement and state equations of the system can be written as:

$$y_t^i = Hx_t + w_t, \quad (\text{C.83})$$

$$x_t = \hat{\rho}x_{t-1} + u_t \quad (\text{C.84})$$

where $y_t^i = [s_t^i, s_t^i]'$, $H = [1, \quad 1]'$, $w_t = [\epsilon_{it}, \quad \nu_t]'$ Let $\Sigma_{t|t-1}^i = \mathbb{E} \left[\left(x_t - x_{t|t-1}^i \right)^2 \right]$ be their prior variance belief and $K_t^v = [K_{1t}, K_{2t}]'$ denote the vector of Kalman gains. The Kalman recursion satisfies the following equations:

$$K_t^v = \Sigma_{t|t-1}^i H' (H \Sigma_{t|t-1}^i H' + R)^{-1}, \quad (\text{C.85})$$

$$\Sigma_{t|t}^i = (1 - K_t^v H) \Sigma_{t|t-1}^i, \quad (\text{C.86})$$

$$\Sigma_{t+1|t}^i = \hat{\rho}^2 \Sigma_{t|t}^i + \sigma_u^2. \quad (\text{C.87})$$

with the matrix R given by the following expression:

$$\begin{bmatrix} \sigma_\epsilon^2 & 0 \\ 0 & \sigma_\nu^2 \end{bmatrix} \quad (\text{C.88})$$

Using equations from the Kalman recursion, the steady-state prior variance $\Sigma_{t|t-1}^i = \Sigma$ is given by the following expression:

$$\Sigma = \frac{\sigma_u^2 (\sigma_\epsilon^2 + \sigma_\nu^2) - (1 - \hat{\rho}^2) \sigma_\nu^2 \sigma_\epsilon^2 + \sqrt{[\sigma_u^2 (\sigma_\epsilon^2 + \sigma_\nu^2) - (1 - \hat{\rho}^2) \sigma_\epsilon^2 \sigma_\nu^2]^2 + 4 (\sigma_\epsilon^2 + \sigma_\nu^2) \sigma_\nu^2 \sigma_u^2 \sigma_\epsilon^2}}{2 (\sigma_\epsilon^2 + \sigma_\nu^2)} \quad (\text{C.89})$$

which is the same as equation (C.58), which σ_ϵ^2 in place of $\hat{\sigma}_\epsilon^2$ and $\hat{\rho}$ in place of ρ . The steady state (distorted) Kalman gains on the public and private signals are then given by the following expressions:

$$K_1 = \frac{\Sigma \sigma_\nu^2}{(\sigma_\epsilon^2 + \sigma_\nu^2) \Sigma + \sigma_\epsilon^2 \sigma_\nu^2}, \quad (\text{C.90})$$

$$K_2 = \frac{\Sigma \sigma_\epsilon^2}{(\sigma_\epsilon^2 + \sigma_\nu^2) \Sigma + \sigma_\epsilon^2 \sigma_\nu^2}. \quad (\text{C.91})$$

The Kalman updating formula at the steady-state Kalman gains gives us the following expression:

$$x_{t|t}^i = x_{t|t-1}^i + K_1(s_t^i - x_{t|t-1}^i) + K_2(s_t - x_{t|t-1}^i) \quad (\text{C.92})$$

$$\Rightarrow \frac{x_{t|t}^i - x_{t|t-1}^i}{K} = x_t - x_{t|t-1}^i + \frac{K_1}{K} \epsilon_{it} + \frac{K_2}{K} \nu_t \quad (\text{C.93})$$

$$\Rightarrow (x_{t|t}^i - x_{t|t-1}^i) \left(\frac{1-K}{K} \right) = x_t - x_{t|t-1}^i + \frac{K_1}{K} \epsilon_{it} + \frac{K_2}{K} \nu_t \quad (\text{C.94})$$

The expression for the individual level coefficient, β_I^h is given by:

$$\beta_I^h = \frac{\text{Cov} \left[\left(x_{t+h}^i - x_{t+h|t}^i \right), \left(x_{t+h|t}^i - x_{t+h|t-1}^i \right) \right]}{\text{Var}(x_{t+h|t}^i - x_{t+h|t-1}^i)} \quad (\text{C.95})$$

$$= \frac{\text{Cov} \left[\rho^h x_t - \hat{\rho}^h x_{t|t}^i, \hat{\rho}^h (x_{t|t}^i - x_{t|t-1}^i) \right]}{\hat{\rho}^{2h} \text{Var} \left(x_{t|t}^i - x_{t|t-1}^i \right)} \quad (\text{C.96})$$

$$= \frac{\text{Cov} \left[(\rho^h - \hat{\rho}^h + \hat{\rho}^h) x_t - \hat{\rho}^h x_{t|t}^i, \hat{\rho}^h (x_{t|t}^i - x_{t|t-1}^i) \right]}{\hat{\rho}^{2h} \text{Var} \left(x_{t|t}^i - x_{t|t-1}^i \right)} \quad (\text{C.97})$$

$$= \frac{\text{Cov} \left[(\rho^h - \hat{\rho}^h) x_t, \hat{\rho}^h (x_{t|t}^i - x_{t|t-1}^i) \right]}{\hat{\rho}^{2h} \text{Var} \left(x_{t|t}^i - x_{t|t-1}^i \right)} + \frac{\text{Cov} \left[\hat{\rho}^h (x_t - x_{t|t}^i), \hat{\rho}^h (x_{t|t}^i - x_{t|t-1}^i) \right]}{\hat{\rho}^{2h} \text{Var} \left(x_{t|t}^i - x_{t|t-1}^i \right)} \quad (\text{C.98})$$

$$= \frac{(\rho^h - \hat{\rho}^h) \text{Cov} \left[x_t, x_{t|t}^i - x_{t|t-1}^i \right]}{\hat{\rho}^h \text{Var} \left(x_{t|t}^i - x_{t|t-1}^i \right)} + \frac{\text{Cov} \left[(x_t - x_{t|t}^i), x_{t|t}^i - x_{t|t-1}^i \right]}{\text{Var} \left(x_{t|t}^i - x_{t|t-1}^i \right)} \quad (\text{C.99})$$

Using equation (C.94), it can be shown that

$$= \frac{\text{Cov} \left(x_t - x_{t|t}^i, x_{t|t}^i - x_{t-1|t-1}^i \right)}{\text{Var} \left(x_{t|t}^i - x_{t-1|t-1}^i \right)} \quad (\text{C.100})$$

$$= \frac{1-K}{K} - \frac{K_1^2 \sigma_\epsilon^2}{K V_{ind}} - \frac{K_2^2 \sigma_\nu^2}{K V_{ind}}. \quad (\text{C.101})$$

where $V_{ind} = \text{Var} \left(x_{t|t}^i - x_{t-1|t-1}^i \right)$. Substituting the above expression into equation (C.99)

$$\beta_I^h = \frac{1-K}{K} - \frac{K_1^2 \sigma_\epsilon^2}{K V_{ind}} - \frac{K_2^2 \sigma_\nu^2}{K V_{ind}} + \frac{(\rho^h - \hat{\rho}^h) C_{x,Ind}}{\hat{\rho}^h V_{ind}}. \quad (\text{C.102})$$

where $C_{x,ind} = \text{Cov}(x_t, x_{t|t}^i - x_{t|t-1}^i)$. We now have to calculate V_{ind} and $C_{x,ind}$. Using equation (C.92), it can be shown that²⁵

$$C_{x,ind} = \text{Cov}(x_t, x_{t|t}^i - x_{t|t-1}^i) = \text{Cov}(x_t, K(x_t - x_{t|t-1}^i) + K_1 \epsilon_{it} + K_2 \nu_t) \quad (\text{C.103})$$

$$= \text{Cov}(x_t - x_{t|t-1}^i + x_{t|t-1}^i, K(x_t - x_{t|t-1}^i)) \quad (\text{C.104})$$

$$= K \text{Var}(x_t - x_{t|t-1}^i) + K \text{Cov}(x_{t|t-1}^i, x_t - x_{t|t-1}^i) \quad (\text{C.105})$$

Furthermore, using equation (C.92), it can be shown that

$$V_{ind} = \text{Var}(x_{t|t}^i - x_{t|t-1}^i) = K^2 \text{Var}(x_t - x_{t|t-1}^i) + K_1^2 \sigma_\epsilon^2 + K_2^2 \sigma_\nu^2 \quad (\text{C.106})$$

In order to complete the calculation, we need expressions for $\text{Var}(x_t - x_{t|t-1}^i)$ and $\text{Cov}(x_{t|t-1}^i, x_t - x_{t|t-1}^i)$. Let $a_t^i = x_t - x_{t|t-1}^i$ and $b_t^i = x_{t|t-1}^i$. It can then be shown that:

$$a_{t+1}^i = (\rho - \hat{\rho}K)a_t^i + (\rho - \hat{\rho})b_t^i + u_{t+1} - \hat{\rho}(K_1 \epsilon_{it} + K_2 \nu_t) \quad (\text{C.107})$$

$$b_{t+1}^i = \hat{\rho}b_t^i + \hat{\rho}K a_t^i + \hat{\rho}(K_1 \epsilon_{it} + K_2 \nu_t) \quad (\text{C.108})$$

Written as a VAR, we have that:

$$\begin{bmatrix} a_{t+1}^i \\ b_{t+1}^i \end{bmatrix} = \begin{bmatrix} \rho - \hat{\rho}K & \rho - \hat{\rho} \\ \hat{\rho}K & \hat{\rho} \end{bmatrix} \begin{bmatrix} a_t^i \\ b_t^i \end{bmatrix} + \begin{bmatrix} 1 & -\hat{\rho} \\ 0 & \hat{\rho} \end{bmatrix} \begin{bmatrix} u_{t+1} \\ K_1 \epsilon_{it} + K_2 \nu_t \end{bmatrix} \quad (\text{C.109})$$

Solving for the steady state variance of this system gives us expressions for $\text{Var}(x_t - x_{t|t-1}^i)$ and $\text{Cov}(x_{t|t-1}^i, x_t - x_{t|t-1}^i)$, which completes our calculation of β_I^h .²⁶

Integrating equation (C.94) across individuals and following similar steps as above allows us

²⁵With extrapolative expectations, this calculation is more involved than under overconfidence because, in the true DGP, $\text{Cov}(x_{t|t-1}^i, x_t - x_{t|t-1}^i) \neq 0$. The reason is that forecasters believe the process is more persistent than it actually is, so they systematically overpredict after high values and underpredict after low values. This leads to forecast errors that are negatively related to the forecast itself, making $\text{Cov}(x_{t|t-1}^i, x_t - x_{t|t-1}^i) < 0$

²⁶In order to solve this system for the steady state variance, we call a matlab solver

to write the following expression for β_C^h :

$$\beta_C^h = \frac{\text{Cov}[(x_{t+h} - x_{t+h|t}), (x_{t+h|t} - x_{t+h|t-1})]}{\text{Var}(x_{t+h|t} - x_{t+h|t-1})} \quad (\text{C.110})$$

$$= \frac{1-K}{K} - \frac{K_2^2 \sigma_\nu^2}{K V_{agg}} + \frac{(\rho^h - \hat{\rho}^h)}{\hat{\rho}^h} \frac{C_{x,agg}}{V_{agg}}. \quad (\text{C.111})$$

where $C_{x,agg} = \text{Cov}(x_t, x_{t|t} - x_{t|t-1})$ and $V_{agg} = \text{Var}(x_{t|t} - x_{t|t-1})$. Also, in a similar manner to equations (C.105) and (C.106), we can show that:

$$C_{x,agg} = \text{Cov}(x_t, x_{t|t} - x_{t|t-1}) \quad (\text{C.112})$$

$$= K \text{Var}(x_t - x_{t|t-1}) + K \text{Cov}(x_{t|t-1}, x_t - x_{t|t-1}) \quad (\text{C.113})$$

$$V_{agg} = \text{Var}(x_{t|t} - x_{t|t-1}) \quad (\text{C.114})$$

$$= K^2 \text{Var}(x_t - x_{t|t-1}) + K_1^2 \sigma_\epsilon^2 + K_2^2 \sigma_\nu^2 \quad (\text{C.115})$$

In order to calculate $\text{Var}(x_t - x_{t|t-1})$ and $\text{Cov}(x_t - x_{t|t-1}, x_{t|t-1})$, we write down the 2x2 equation system similar to (C.109). Letting $x_t - x_{t|t-1} = a_t$ and $x_{t|t-1} = b_t$, one can show that:

$$\begin{bmatrix} a_{t+1} \\ b_{t+1} \end{bmatrix} = \begin{bmatrix} \rho - \hat{\rho}K & \rho - \hat{\rho} \\ \hat{\rho}K & \hat{\rho} \end{bmatrix} \begin{bmatrix} a_t \\ b_t \end{bmatrix} + \begin{bmatrix} 1 & -\hat{\rho} \\ 0 & \hat{\rho} \end{bmatrix} \begin{bmatrix} u_{t+1} \\ K_2 \nu_t \end{bmatrix} \quad (\text{C.116})$$

Solving for the steady state variance of this 2x2 system allows us to obtain the expressions needed for our calculation of β_C^h ²⁷. Finally, we also show how to calculate the cross-sectional variance of forecasts. From equation (C.92), we integrate across i to get $x_{t|t}$. Subtracting

²⁷As before, in order to solve this system for the steady state variance, we call a matlab solver

$x_{t|t}$ from $x_{t|t}^i$, we have:

$$x_{t|t}^i - x_{t|t} = (1 - K)(x_{t|t-1}^i - x_{t|t-1}) + K_1 \epsilon_{it} \quad (\text{C.117})$$

$$= (1 - K) \hat{\rho} (x_{t-1|t-1}^i - x_{t-1|t-1}) + K_1 \epsilon_{it} \quad (\text{C.118})$$

$$\Rightarrow \text{Var}(x_{t|t}^i - x_{t|t}) = \frac{K_1^2 \sigma_\epsilon^2}{1 - (1 - K)^2 \hat{\rho}^2} \quad (\text{C.119})$$

$$\Rightarrow \text{Var}(x_{t+h|t}^i - x_{t+h|t}) = \text{Var}(\hat{\rho}^h (x_{t|t}^i - x_{t|t})) \quad (\text{C.120})$$

$$= \hat{\rho}^{2h} \text{Var}(x_{t|t}^i - x_{t|t}) \quad (\text{C.121})$$

In our GMM exercises we target the ratio of the standard deviation of forecasts between treated and non-treated firms to discipline the value of σ_ν^2 .

D Robustness to Alternative DGP Specifications

We report overidentification tests under alternative data-generating-process parameters (ρ, σ_u^2) . The results are quite similar to those from Table (7): the Overconfidence model provides the best fit; the Over-extrapolation model is strongly rejected because it cannot match the non-treated moments; and the Noisy Memory model lies in between-matching the non-treated coefficients reasonably well but failing to reproduce the magnitude of the treated coefficients for plausible values of σ_ν^2 .

D.1 Just-Identified Exercises For Model Intuition

To build intuition for the model-selection results in Section 5.6, we conduct two just-identified exercises. In both exercises, we choose σ_ϵ^2 and the model-specific behavioral parameter to match the non-treated moments, $\beta_C(NT)$ and $\beta_I(NT)$; the calibrated parameter values are reported in Table D.7. The exercises then ask whether each model can use the public-signal variance σ_ν^2 to reproduce the treated moments.

In the first exercise, we choose σ_ν^2 to match the treated consensus coefficient $\beta_C(T)$, leaving the treated individual coefficient $\beta_I(T)$ and the treated/non-treated forecast-dispersion ratio as non-targeted moments. Panel A of Table D.8 shows that the Overconfidence model

Table D.1: Calibrated Parameters, Overidentification Test, $\rho = 0.8$

Parameter	Model		
	Overconfidence	Noisy Memory	Extrapolation
ρ (Fix)	0.80	0.80	0.80
σ_u^2 (Fix)	0.42	0.42	0.42
σ_ϵ^2	1.65	1.11	2.27
$\hat{\sigma}_\epsilon^2$	0.78	—	—
λ	—	0.28	—
$\hat{\rho}$	—	—	0.96
σ_ν^2 (Treated)	1.19	1.54	3.80

Table (D.1) reports the calibrated parameters from our overidentification test when we set the value of $\rho = 0.8$.

Table D.2: Overidentification Test, Results: $\rho = 0.8$

Moment	Data	Model		
		Overconfidence	Noisy Memory	Extrapolation
β_C (T)	0.43	0.33	0.02	0.20
β_I (T)	-0.15	-0.18	-0.25	-0.15
β_C (NT)	1.25	1.19	1.06	1.52
β_I (NT)	-0.28	-0.29	-0.28	-0.18
SD ratio (T/NT)	0.70	0.71	0.73	0.71
J-statistic		0.25	4.17	9.46
p-value		0.88	0.12	<0.01

Table (D.2) reports the results of our overidentification test when $\rho = 0.8$.

Table D.3: Calibrated Parameters, Overidentification Test, $\rho = 0.9$

Parameter	Model		
	Overconfidence	Noisy Memory	Extrapolation
ρ (Fix)	0.90	0.90	0.90
σ_u^2 (Fix)	0.42	0.42	0.42
σ_ϵ^2	2.15	1.39	2.54
$\hat{\sigma}_\epsilon^2$	0.94	—	—
λ	—	0.71	—
$\hat{\rho}$	—	—	0.99
σ_ν^2 (Treated)	1.50	2.09	4.09

Table (D.3) reports the calibrated parameters from our overidentification test when we set the value of $\rho = 0.9$.

Table D.4: Overidentification Test, Results: $\rho = 0.9$

Moment	Data	Model		
		Overconfidence	Noisy Memory	Extrapolation
β_C (T)	0.43	0.34	0.05	0.30
β_I (T)	-0.15	-0.19	-0.25	-0.08
β_C (NT)	1.25	1.22	1.10	1.80
β_I (NT)	-0.28	-0.28	-0.28	-0.10
SD ratio (T/NT)	0.70	0.71	0.73	0.70
J-statistic		0.18	3.30	23.28
p-value		0.92	0.19	<0.01

Table (D.4) reports the results of our overidentification test when $\rho = 0.9$.

Table D.5: Calibrated Parameters, Overidentification Test, $\sigma_u^2 = 0.7$

Parameter	Model		
	Overconfidence	Noisy Memory	Extrapolation
ρ (Fix)	0.85	0.85	0.85
σ_u^2 (Fix)	0.70	0.70	0.70
σ_ϵ^2	3.04	2.00	3.99
$\hat{\sigma}_\epsilon^2$	1.40	—	—
λ	—	0.50	—
$\hat{\rho}$	—	—	0.97
σ_ν^2 (Treated)	2.17	2.88	6.57

Table (D.5) reports the calibrated parameters from our overidentification test when we set the value of $\sigma_u^2 = 0.7$.

Table D.6: Overidentification Test, Results: $\sigma_u^2 = 0.7$

Moment	Data	Model		
		Overconfidence	Noisy Memory	Extrapolation
β_C (T)	0.43	0.33	0.03	0.25
β_I (T)	-0.15	-0.18	-0.25	-0.12
β_C (NT)	1.25	1.21	1.08	1.65
β_I (NT)	-0.28	-0.28	-0.28	-0.15
SD ratio (T/NT)	0.70	0.71	0.73	0.71
J-statistic		0.19	3.78	14.57
p-value		0.91	0.15	<0.01

Table (D.6) reports the results of our overidentification test when $\sigma_u^2 = 0.7$.

performs best on the non-targeted individual moment: it implies $\beta_I(T) = -0.20$, close to the empirical value of -0.15 . By contrast, the Noisy Memory model leaves $\beta_I(T)$ essentially unchanged at -0.27 , while the Over-extrapolation model cannot match the non-treated individual coefficient in the first place.

In the second exercise, we instead choose σ_ν^2 to match the treated individual coefficient $\beta_I(T)$, leaving $\beta_C(T)$ and forecast dispersion untargeted. This exercise highlights the main weakness of the Noisy Memory model. Because β_I responds only weakly to public-signal precision, matching the treated β_I requires an extremely precise public signal. That signal makes treated firms load too heavily on the same common information, causing forecast dispersion to collapse and pushing the treated consensus coefficient below zero. In Panel B of Table D.8, the Noisy Memory model implies $\beta_C(T) = -0.11$ and a treated/non-treated dispersion ratio of 0.29, far from the empirical values of 0.43 and 0.70. The Overconfidence model does not face the same tradeoff because the public signal directly reduces the relative weight placed on overconfident private information.

Figure D.1 illustrates the same logic by plotting β_C and β_I as public-signal precision varies across models; Figure D.2 overlays the three specifications on common axes, and Figure D.3 plots the corresponding Kalman gains on the private and public signals.

Table D.7: Calibration Details: Just Identified GMM Exercises

Parameter	Model		
	Overconfidence	Noisy Memory	Extrapolation
ρ (Fix)	0.85	0.85	0.85
σ_u^2 (Fix)	0.42	0.42	0.42
σ_ϵ^2	1.89	1.39	1.59
$\hat{\sigma}_\epsilon^2$	0.88	—	—
λ	—	0.52	—
$\hat{\rho}$	—	—	0.97
σ_ν^2 (Treated, Exercise 1)	1.88	7.00	5.30
σ_ν^2 (Treated, Exercise 2)	0.92	0.26	—

Table (D.7) reports the parameters calibrated in our just-identified GMM Exercises.

Table D.8: Just-Identified GMM Exercises: Results

Panel A: Just-identified GMM, Exercise 1 (σ_ν^2 targets β_C (T))					
Moments	Targeted	Data	Noisy Memory	Overconfidence	Extrapolation
β_C (NT)	Yes	1.25	1.25	1.25	1.26
β_I (NT)	Yes	-0.28	-0.28	-0.28	-0.13
β_C (T)	Yes	0.43	0.43	0.43	0.43
β_I (T)	No	-0.15	-0.27	-0.20	-0.11
SD (T/NT)	No	0.70	0.91	0.77	0.82

Panel B: Just-identified GMM, Exercise 2 (σ_ν^2 targets β_I (T))				
Moment	Targeted	Data	Noisy Memory	Overconfidence
β_C (NT)	Yes	1.25	1.25	1.25
β_C (T)	No	0.43	-0.11	0.24
β_I (NT)	Yes	-0.28	-0.28	-0.28
β_I (T)	Yes	-0.15	-0.15	-0.15
SD (T/NT)	No	0.70	0.29	0.62

Panel A calibrates σ_ν^2 to match β_C (treated), leaving treated β_I non-targeted. Panel B calibrates σ_ν^2 to match β_I (treated), leaving treated β_C non-targeted. We exclude the Over-extrapolation model from the second exercise because, under our chosen DGP, there is no admissible parameter vector $(\sigma_\epsilon^2, \hat{\rho}, \sigma_\nu^2)$ that can simultaneously match β_I (NT) and β_I (T).

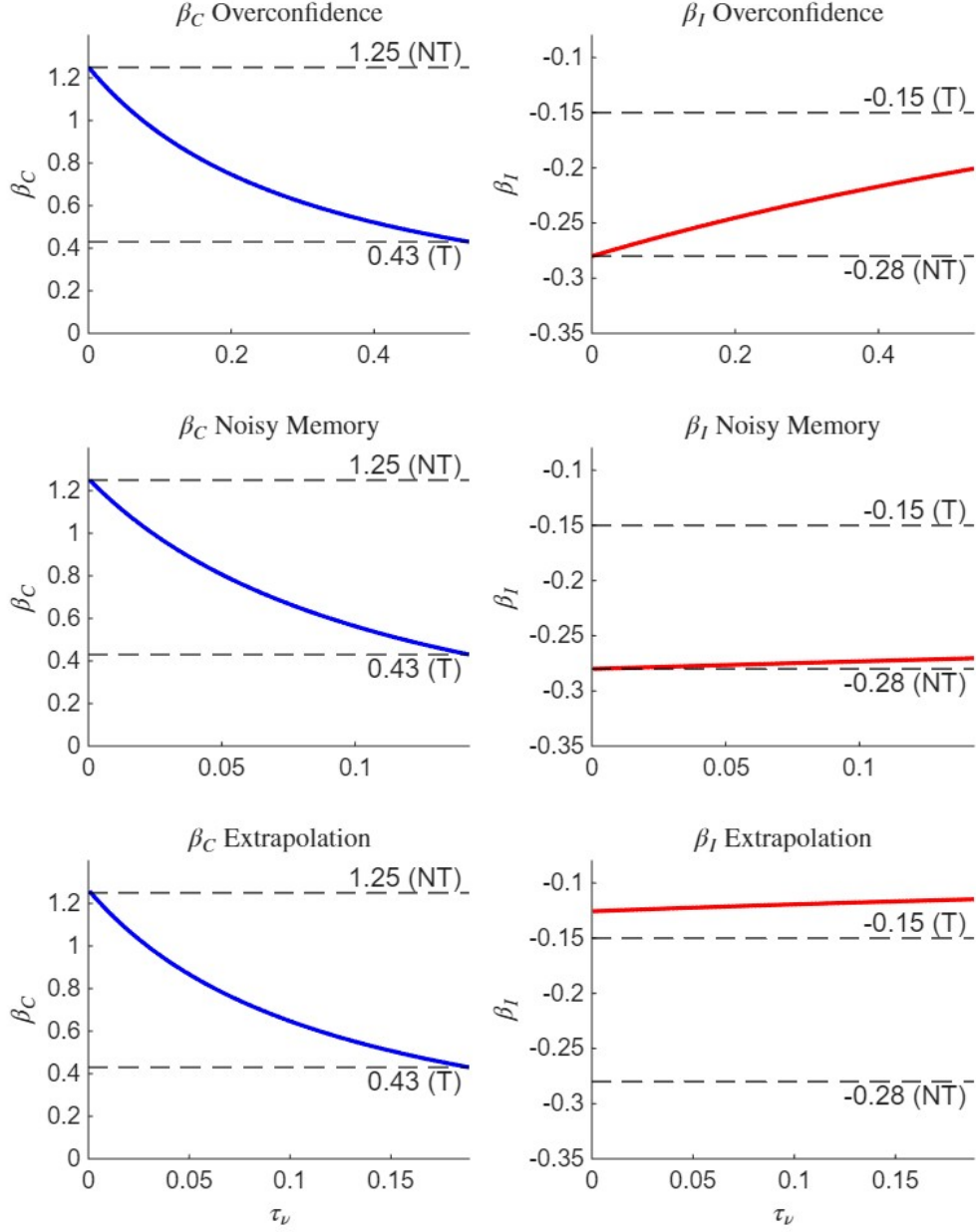
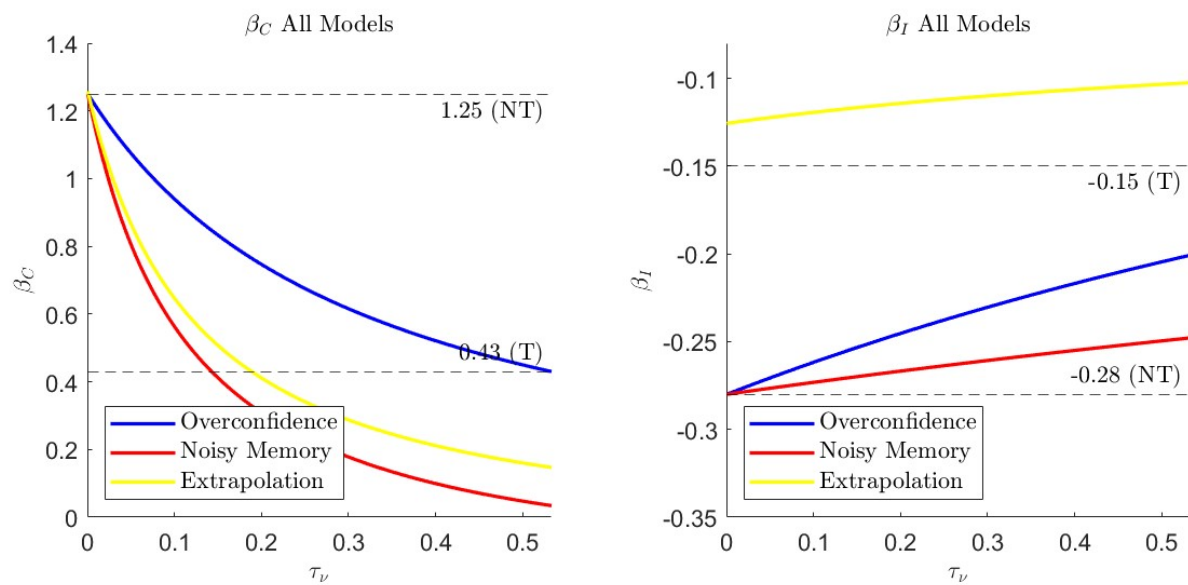


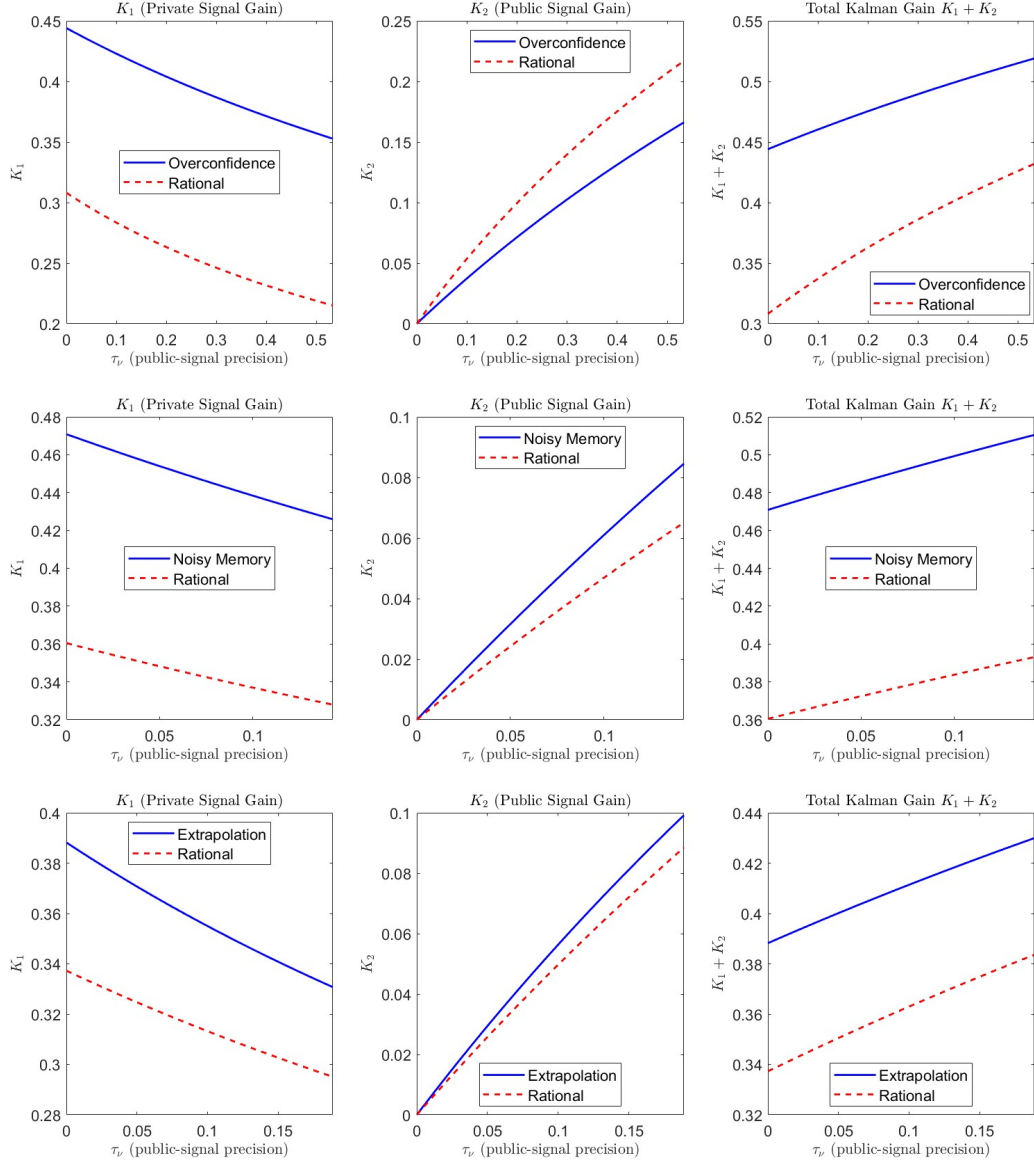
Figure D.1: Just-identified intuition exercise: public-signal precision and forecast-error predictability. The figure plots the consensus coefficient β_C and individual coefficient β_I as public-signal precision varies across the three model specifications.

Figure D.2: β_C, β_I as a function of public signal precision τ_ν under the three model specifications:



The figure overlays the three model specifications, plotting how increases in public-signal precision affect β_C and β_I on common axes.

Figure D.3: Kalman gain on Private and Public Signals Across Models.



The figure shows the Kalman gains on Private and Public Signals (K_1 and K_2 respectively) as a function of public signal precision

E Derivations and Additional Results for Section 6

Throughout this appendix we use the composite parameters of Angeletos and Huo (2021): $\beta \equiv \chi\theta$, $\gamma \equiv \chi(1 - \theta)$, and $\delta \equiv \beta + \gamma = \chi$. (These symbols are appendix-local; the regression coefficients of the main text always carry subscripts, β_C and β_I . We denote the Wiener–Hopf root of Angeletos and Huo (2021) by μ rather than by their λ , which Section 5.1 reserves for the memory parameter.) The fundamental is (6.2) with $\sigma_u^2 = 1$.

The slope $\kappa = 0.17$ of the main text is anchored as follows. Under the Calvo frequency conversion implied by our θ – the slope scales by $(1 + \chi_q\theta_q)(1 + \theta_q)/\theta_q \approx 4.25$ between the quarterly and six-month frequencies – $\kappa = 0.17$ corresponds to a quarterly slope on real marginal cost of about 0.04, in line with the GMM estimates of Galí and Gertler (1999), who measure marginal cost by the same labor-share proxy we use for ρ . Equivalently, holding the Calvo no-reset probability at its micro-consistent value, $\kappa = 0.17$ corresponds to a real-rigidity factor of roughly 0.15 relative to the textbook slope $(1 - \chi\theta)(1 - \theta)/\theta$; real rigidities scale only the marginal-cost loading and leave the information-friction calibration unchanged.

E.1 The consensus and individual coefficients in the Angeletos–Huo economy

The equilibrium law of motion (6.5) and the wedge formulas (6.7) restate Proposition 2 and Online Appendix eqs. A5–A6 of A&H, with $\delta = \chi$; the root ϑ solves the cubic given in Appendix E.3 below, with $\tau_\nu = 0$ in the private-information economy of the non-treated calibration. The scale A cancels from every moment ratio used in the calibration.

The consensus error-on-revision coefficient in this economy has the closed form of A&H (their eq. 24):

$$K_{CG} = \mu \frac{\vartheta + \rho - \rho\vartheta(\mu + \vartheta) - \rho\mu\vartheta(1 - \mu\vartheta)}{(\rho - \mu)(1 - \mu\vartheta)(\rho + \vartheta - \mu\rho\vartheta)}, \quad \mu = \frac{1}{2} \left(A_\mu - \sqrt{A_\mu^2 - 4} \right), \quad (\text{E.1})$$

where $A_\mu = \rho + \rho^{-1}(1 + \sigma_u^2/\hat{\sigma}_\epsilon^2)$ and μ is the Wiener–Hopf (Kalman) root governing first-order learning. K_{CG} is monotonically increasing in the perceived noise, so $\beta_C(NT) = 1.25$ identifies $\hat{\sigma}_\epsilon^2$ uniquely. By Proposition 7 of A&H, (6.5), (6.7), and (E.1) depend on the private-signal noise only through its *perceived* variance.

We compute the individual coefficient β_I from the same state space at the firm level: each firm filters with the steady-state Kalman gain implied by its *perceived* noise, while its signals carry the *true* noise; the stationary covariance of the joint system of fundamentals, individual beliefs, and lagged beliefs delivers $\beta_I = \text{Cov}(FE_i, REV_i)/\text{Var}(REV_i)$. Under rational perception $\beta_I = 0$ by orthogonality; overconfidence makes it negative, and $\beta_I(NT) = -0.28$ identifies σ_ϵ^2 given $\hat{\sigma}_\epsilon^2$.

E.2 The fixed-outcome identification of the public-signal precision

The identification of τ_ν^{PE} solves a pure forecasting problem in which we freeze the outcome process at the non-treated equilibrium. Stacking the state as $\zeta_t = (mc_t, q_t)'$, the state-space representation of the fixed-outcome PE identification is

$$\zeta_t = T \zeta_{t-1} + B u_t, \quad T = \begin{pmatrix} \rho & 0 \\ \rho & \vartheta_{NT} \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \pi_t^{NT} = A_{NT} q_t, \quad (\text{E.2})$$

the equilibrium law (6.5) evaluated at the non-treated root ϑ_{NT} from Appendix E.1. Treated firms observe $y_{i,t} = (s_t^i, s_t)' = H \zeta_t + (\epsilon_{it}, \nu_t)'$ with $H = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ and perceived noise covariance $\text{diag}(\hat{\sigma}_\epsilon^2, 1/\tau_\nu)$, and update with the implied steady-state Kalman gain K . Averaging the belief recursion across the continuum integrates out the idiosyncratic noise; the common noise ν_t survives:

$$\bar{m}_t = (I - KH)T \bar{m}_{t-1} + KHT \zeta_{t-1} + KHB u_t + KD \nu_t, \quad (\text{E.3})$$

where $\bar{m}_t$ is the average posterior mean of ζ_t and $D = (0, 1)'$ selects the public-noise loading. The stacked vector $X_t = (\zeta_t', \bar{m}_t', \bar{m}_{t-1}')'$ follows a first-order linear system whose stationary covariance Σ_X solves a discrete Lyapunov equation. With $e_2 = (0, 1)$, the consensus forecast error and revision for the one-period-ahead outcome are the linear functionals

$$FE_t = A_{NT} e_2 T (\zeta_t - \bar{m}_t), \quad REV_t = A_{NT} e_2 T \bar{m}_t - A_{NT} e_2 T^2 \bar{m}_{t-1}, \quad (\text{E.4})$$

where FE_t omits the unforecastable innovation $A_{NT} u_{t+1}$, which is orthogonal to REV_t and hence immaterial below. The moments in (6.10) are therefore quadratic forms in Σ_X , and

$K_{CG}^{PE}(\tau_\nu)$ follows. The scale A_{NT} cancels from (6.10). At $\tau_\nu = 0$ the system reduces to the private-information problem and (6.10) reproduces the closed form (E.1) exactly; for $\tau_\nu > 0$, K_{CG}^{PE} is monotonically decreasing in τ_ν on the calibration grid, so $\beta_C(T) = 0.43$ identifies τ_ν^{PE} uniquely. Throughout, τ_ν enters only the observation system and the Kalman gain – never the transition matrix T : the public signal moves beliefs, not the outcome.

E.3 The public-information equilibrium

The counterfactual of Section 6.3 re-solves the equilibrium when the signal s_t is common to all price setters. Proposition 12 in the Online Appendix of A&H characterizes this economy. (They state the proposition under rational perception of the private noise; the argument behind their Proposition 7 extends to this economy, and we accordingly write the cubic with the perceived precision $\hat{\sigma}_\epsilon^{-2}$. Operationally, this is the same invariance imposed in the calibration: true private noise enters only the individual-moment calculation, while the public-signal cubic and the macro wedges use perceived private precision.) The friction root is $\vartheta = 1/z_{\text{out}}$, where $z_{\text{out}} > 1$ is the outside root of

$$\begin{aligned} z^3 - a_2 z^2 + a_1 z - a_0 &= 0, & a_2 &= \rho + \frac{1}{\rho} + \frac{1}{\rho} P + \beta, \\ a_1 &= 1 + \beta\left(\rho + \frac{1}{\rho}\right) + \frac{1}{\rho}(\delta - \gamma\psi)P, & a_0 &= \beta, \end{aligned} \tag{E.5}$$

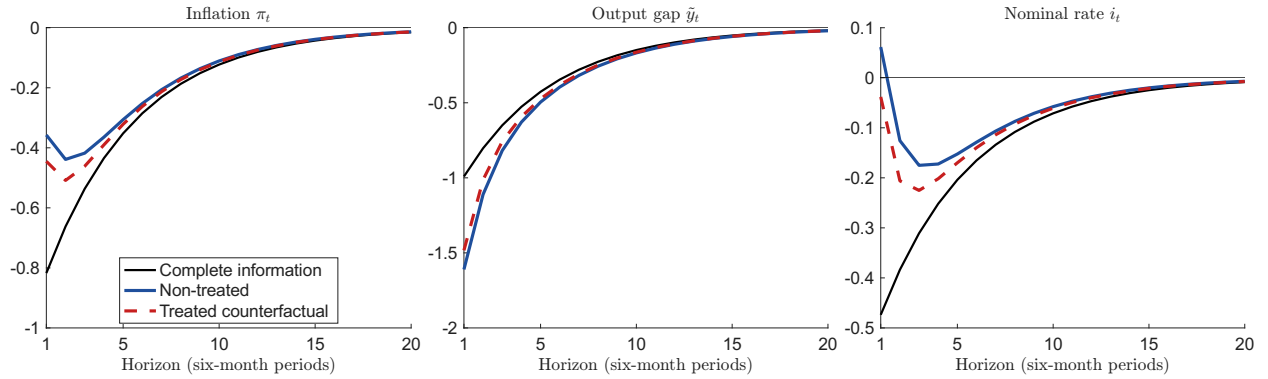
where $P = \sigma_u^2(\hat{\sigma}_\epsilon^{-2} + \tau_\nu)$ is the total perceived precision relative to the fundamental and $\psi = \tau_\nu/(\hat{\sigma}_\epsilon^{-2} + \tau_\nu)$ is the public-signal share of perceived precision, common to all firms in this economy; its value at τ_ν^{PE} is the ψ^{PE} of (6.12). The private-information economy of Appendix E.1 is the special case $\tau_\nu = 0$ ($P = \sigma_u^2/\hat{\sigma}_\epsilon^2$, $\psi = 0$). Two features of (E.5) organize the economics: total precision P captures the partial-equilibrium effect of better information, while the substitution $\delta \mapsto \delta - \gamma\psi$ captures the effect of *publicness* alone, which operates only through the GE weight γ . Proposition 12 also implies that the equilibrium outcome acquires a residual component driven by the common noise ν_t ; the wedges and the impulse responses to fundamentals in the main text describe the fundamental component. Given the root, the wedges follow from (6.7). At $(\hat{\sigma}_\epsilon^2, \tau_\nu^{PE}) = (2.16, 0.21)$ the cubic delivers $\vartheta_T = 0.434$ and the treated-counterfactual wedges of Table 9. The treated diagnostics of Section 6.2 – the indi-

vidual coefficient and the forecast-dispersion ratio – are moments of this economy evaluated at τ_ν^{PE} : public-counterfactual diagnostics, not fixed-outcome identification moments.

E.4 Responses to a monetary-policy shock

Figure E.1 repeats the exercise of Figure 3 for a one-standard-deviation monetary-policy shock (an AR(1) disturbance in the Taylor rule with six-month persistence 0.81). The message mirrors the cost-push case: the information friction cuts the impact response of inflation by more than half and delays its trough. Information provision again moves all three responses toward the FIRE benchmark, raising the impact response of inflation by about 25%.

Figure E.1: Responses to a monetary-policy shock under the three information regimes



Conventions as in Figure 3.

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